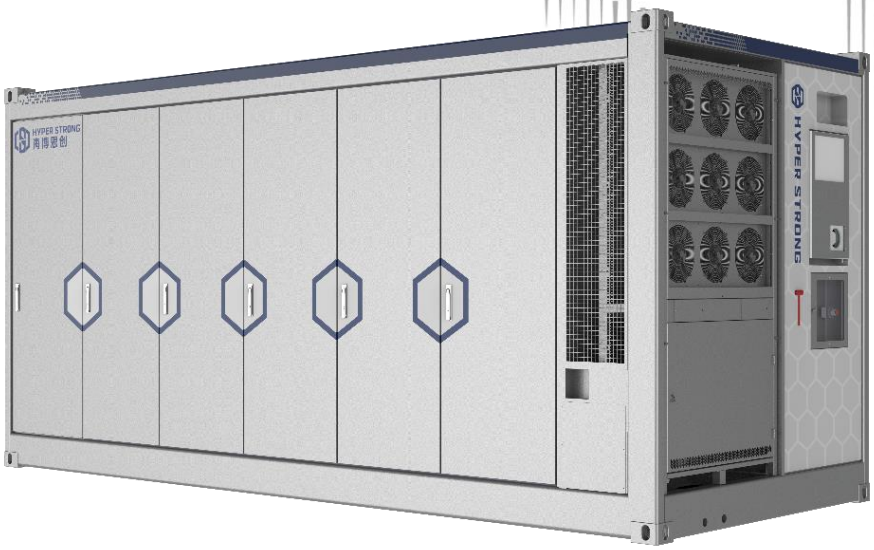
HYPER STRONG



User Manual

20Ft Battery Energy Storage System

HSL3C7001-05015EU/US HSL3C7001-04597EU/US

HSL3C7001-04179EU/US HSL3C7001-03761EU/US

HSL3C7001-03343EU/US

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PREFACE

Manual Application

This manual is prepared according to the specifications of the initial edition of the liquid-cooled container models:

HSL3C7001-05015EU/US

HSL3C7001-04597EU/US

HSL3C7001-04179EU/US

HSL3C7001-03761EU/US

HSL3C7001-03343EU/US

The technical specifications of this product are subject to updates, and this manual may be revised without prior notice. This manual applies to the energy storage battery system, focusing on product operation and functionality during transportation, assembly, installation, and commissioning. Pictures, data, and instructions herein are for reference purposes, shall not be considered as the basis for order acceptance.

Target Audience

This manual is intended for operators (customers), qualified electrical engineers, operations and after-sales personnel. Please follow the instructions provided in this manual to operate the system. Failure to do so may cause equipment damage, personnel injury, property loss.

Retention

This manual contains crucial information. Please

1. Read it carefully before operating and maintaining the system.
2. Operate the system in accordance with the instructions to avoid the risk of equipment damage, injury to personnel, or property loss.
3. Ensure the manual is properly maintained and accessible by operators and maintenance personnel.

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Revision History

Document Revision History

Version	Description of Revision	Author	Auditor	Approver	Date
V1.0	New release	Jia Tian	Heng Yang	Zhenhua Yang	2024/05/14
V1.1	1. Add some certification instructions. 2. Correct some product specifications. 3. Revise the format and layout. 4. Optimize FSS description. 5. According to suggestions from certification organization, add some contents including safety signs, product parameters, cable specifications, wiring protection, and contact information, etc.	Han Zhang and Dashan Zhang	Heng Yang	Zhenhua Yang	2024/08/26
V1.2	1.Add some contents for AS certific ation and update structure picture. 2.Fixed some formatting errors. 3.Fixed product layout distance.	Jia Tian and Han Zhang	Heng Yang	Zhenhua Yang	2024/10/14
V1.3	1.Professionalized the entire content of the document. 2.Modified ambiguous content for clarity. 3.Enhanced the document's formatting.	Han Zhang and jin Zhou	Heng Yang	Zhenhua Yang	2024/12/23
V1.4	1.Update the word abbreviations. 2.Update the pictures and descriptions of related specialties.	Guoyao Wu and other engineers	Guoyao Wu		2025/08/25[i]

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1 Symbol Conventions

The symbols that may be found in this document are defined as follows:

Symbol	Description
DANGER	Indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
WARNING	Indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.
CAUTION	Indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.
NOTICE	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
NOTE	Supplements the important information in the main text. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

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Signe	Décrire
	Indique un danger avec un niveau de risque élevé qui, s'il n'est pas évité, entraînera la mort ou des blessures graves.
	Indique un danger avec un niveau de risque moyen qui, s'il n'est pas évité, pourrait entraîner la mort ou des blessures graves.
	Indique un danger avec un faible niveau de risque qui, s'il n'est pas évité, pourrait entraîner des blessures mineures ou modérées.
	Indique une situation potentiellement dangereuse qui, si elle n'est pas évitée, pourrait entraîner des dommages matériels, une perte de données, une détérioration des performances ou des résultats imprévus. La notification est utilisée pour traiter des pratiques qui ne sont pas liées à des blessures corporelles.
	Complète les informations importantes dans le texte principal. Le commentaire est utilisé pour traiter les informations qui ne sont pas liées aux blessures personnelles, aux dommages matériels et à la détérioration de l'environnement.

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2 Abbreviations

Complete Designation	Abbreviations
System Control and Distribution Cabinet	SCDC
Battery Array Management Unit	BAMU
Battery Cluster Management Unit	BCMU
Battery Energy Storage System	BESS
Battery Module Management Unit	BMU
DC/DC Converter	DC/DC
Energy Management System	EMS
Energy Storage System	ESS
Fire Suppression System	FSS
Human Machine Interface	HMI
Heating, Ventilating and Air Conditioning	HVAC
Local Controller or Control room	LC
Battery Module	PACK
Power Conversion System	PCS
Power Distribution Unit	PDU
Battery Cluster	RACK
SCADA (Supervisory Control And Data Acquisition) System	SCADA
Switching Mode Power Supply	SMPS
State Of Charge	SOC
State Of Energy	SOE
State Of Health	SOH

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3 Safety Instructions

3.1 Safety Signs

Safety signs on the energy storage system equipment contain essential information for safe operations. Manual removal or damage is strictly prohibited.

Table 3.1 shows common safety signs.

Signs	Explanation
	It indicates that the area contains fire-hazardous materials, and pyrotechnics are strictly prohibited in this location.
	It indicates that the equipment in the area shall not be disassembled.
	It indicates that the area contains fire-hazardous materials, and smoking is prohibited in this location.
	It indicates that the device poses a danger of entrapment of body parts and that inserting body parts into the machine is prohibited.
	It indicates that the area prohibits the stacking of items.
	It indicates that all personnel operating the equipment must wear protective gloves to prevent electric shock or static electricity.
	It indicates that the equipment needs to be grounded. The equipment must be grounded through the grounding point, ensuring regular and safe operation.
	It indicates an immediate hazardous situation that, if not avoided, will result in serious personal injury or death.
	It indicates a flammable substance that, if not avoided, could result in severe injury or death.
	It is used to convey safety warning information about equipment or the environment. It may lead to equipment damage, data loss, performance degradation, or other unpredictable results if not avoided.
	It indicates that cables have been placed in this area and you need to be aware of the cables underneath your feet when servicing or passing through the site to prevent tripping.

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	It indicates a risk of electric shock from this equipment or device. Operators should take care while operating this equipment or device to avoid electrocution.
	It indicates that the machinery of the equipment may pose a risk of injury and that care should be taken when operating the equipment to prevent injury from the machinery.
	Do not dispose of this product as household waste.
	Read the instructions before performing any operation on the product.
	It is required to wear goggles.
	Ear protection must be worn.
	In case of contact with the eyes, rinse the eyes immediately with running water or normal saline and seek medical help in time.
	Live parts! Please do not touch them until 5 minutes after disconnecting from the power sources.
	Emergency Exit: It indicates an emergency safety exit in the direction of the arrow of movement and that you can escape in the order given in the event of an emergency danger.

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Signes	Explication
	Il indique que la zone contient des matières dangereuses pour le feu et que la pyrotechnie est strictement interdite à cet endroit.
	Il indique que l'équipement dans la zone ne doit pas être démonté.
	Il indique que la zone contient des matières dangereuses pour les incendies et qu'il est interdit de fumer à cet endroit.
	Il indique que l'appareil présente un risque de coincement de parties ducorps et qu'il est interdit d'insérer des parties du corps dans la machine.
	Il indique que la zone interdit l'empilement d'objets.
	Il indique que tout le personnel utilisant l'équipement doit porter des gants de protection pour éviter les chocs électriques ou l'électricité statique.
	Il indique que l'équipement doit être mis à la terre. L'équipement doit être mis à la terre à travers le point de mise à la terre, ce qui garantit un fonctionnement régulier et sûr.
	Il indique une situation dangereuse immédiate qui, si elle n'est pas évitée, entraînera des blessures graves ou la mort.
	Il s'agit d'une substance inflammable qui, si elle n'est pas évitée, pourrait entraîner des blessures graves ou la mort.
	Il est utilisé pour transmettre des informations d'avertissement de sécurité sur l'équipement ou l'environnement. Cela peut entraîner des dommages matériels, une perte de données, une dégradation des performances ou d'autres résultats imprévisibles s'il n'est pas évité.
	Cela indique que des câbles ont été placés dans cette zone et que vous devez être conscient des câbles sous vos pieds lors de l'entretien ou du passage sur le site pour éviter de trébucher.
	Il indique un risque de choc électrique provenant de cet équipement ou appareil. Les opérateurs doivent faire attention lors de l'utilisation de cet équipement ou appareil pour éviter l'électrocution.
	Il indique que la machinerie de l'équipement peut présenter un risque de blessure et qu'il faut prendre des précautions lors de l'utilisation de l'équipement pour éviter les blessures causées par la machine.

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	Ne jetez pas ce produit avec les ordures ménagères.
	Lisez les instructions avant d'effectuer toute opération sur le produit.
	Il est obligatoire de porter des lunettes.
	Le porte de cache-oreilles est obligatoire.
	En cas de contact avec les yeux, rincer immédiatement les yeux avec de l'eau courante ou du sérum physiologique et consulter un médecin à temps.
	Pièces sous tension! Veuillez ne pas les toucher jusqu'à 5 minutes après les avoir débranchés des sources d'alimentation.
	Sortie de secours : Il indique une sortie de secours de sécurité dans le sens de la flèche de mouvement et que vous pouvez vous échapper dans l'ordre donné en cas de danger d'urgence.

3.2 General Safety Precautions

The safety precautions outlined in this document provide essential guidance for selecting measuring and test equipment during the installation, operation, and maintenance of the equipment.

Statement

Before transporting, storing, installing, operating, using, or maintaining the equipment, carefully read this document, adhere strictly to the instructions provided, and follow all the safety instructions indicated on the equipment and within this document.

For the purpose of this document:

"Equipment" refers to the products, software, components, spare parts, and/or services related to this document.

"The Company" refers to the manufacturer (producer), seller, and/or service provider of the equipment.

"You" refers to the entity that transports, stores, installs, operates, uses, and/or maintains the equipment. The Danger, Warning, Caution, and Notice statements included in this document are supplementary to, but do not encompass, all necessary safety precautions.

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Operators must also comply with relevant international, national, and regional safety standards.

The Company shall not be liable for any consequences resulting from violations of safety requirements or safety standards concerning the design, production, and usage of the equipment.

The equipment must be operated in an environment that meets the design specifications. Use in non-compliant environments may result in faulty, malfunctioning, or damaged, which are not covered under warranty. The Company shall not be liable for any property loss, personal injury, or even death caused thereby.

Ensure compliance with all applicable laws, regulations, standards, and specifications during transportation, storage, installation, operation, use, and maintenance.

Do not perform reverse engineering, decomplication, disassembly, adaptation, implantation, or other derivative operations on the equipment software.

Do not study the internal implementation logic of the equipment, obtain the source code of the equipment software, violate intellectual property rights, or disclose any of the performance test results of the equipment software.

During the storage phase of the energy storage system, the customer should implement safety measures to mitigate unforeseen circumstances.

The Company shall not be liable for any of the following circumstances or their consequences:

- Equipment damage caused force majeure, including but not limited to earthquakes, floods, volcanic eruptions, debris flows, lightning strikes, fires, wars, armed conflicts, typhoons, hurricanes, tornadoes, and other extreme weather conditions.
- Operation of equipment beyond the conditions specified in this document.
- Installation or usage in environments that do not comply with international, national, or regional standards.
- Failure to follow the operation instructions, operation and maintenance manual, or safety precautions on the product and in the document.
- Unauthorized removal, modification of the product, or alteration of software code.
- Equipment damage resulting from improper storage conditions that deviate from the requirements specified in the product document.
- Failure to prepare materials and tools that comply with local laws, regulations, and related standards.
- Negligence, gross negligence, intentional breach, improper operations, or other third-party actions causing equipment damage.
- Unauthorized change to the application of battery energy storage system.

- Battery stored beyond the maximum storage period.
- Failure to charge promptly, resulting in capacity loss or irreversible damage to the battery.
- Installation or operation by unqualified personnel.
- Mixing the batteries supplied by the Company with other batteries, including mixing with different brands or batteries with incompatible capacities, causing accelerated capacity degradation.

Safety Precautions

- During installation, overhaul, or maintenance, place clear warning signs or safety caution tape near the operating area. Prominent signs must also be displayed at the front and rear switches of the energy storage system to prevent accidental closures.
- Technicians responsible for installation and maintenance must be professionally trained, fully understand all safety precautions, be familiar with the composition and working principle, and master correct operation methods.
- Only qualified professionals are authorized to perform equipment removal and overhaul operations.
- Replacement and modification of equipment or components (including software) must be carried out by personnel authorized by HyperStrong.
- Ensure the escape routes remain fully accessible during the maintenance and overhaul activities.
- After completing maintenance, overhaul, and other operations, remove all tools and restore the site to its original state for safe and regular equipment operation.
- During the storage, must establish safety measures to prevent unforeseen incidents.

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3.3 Personal Safety

DANGER	Please wear protection gear before operations to prevent electric shocks or burns.
DANGER	Equipment operates in high voltage. Any accidental contact may result in a fatal shock hazard. It is essential to have at least two personnel present during maintenance or operation for safety.
DANGER	When the equipment is powered on, it is strictly prohibited to move, open, or dismantle the electric equipment container door or touch any internal components. Before performing maintenance operations, ensure the equipment has been safely powered off.
DANGER	Avoid contact cords or operating equipment during storms.

DANGER	Veuillez porter un équipement de protection avant les opérations pour éviter les chocs électriques ou les brûlures.
DANGER	L'équipement est à haute tension. Tout contact accidentel peut entraîner un risque d'électrocution mortelle. Il est essentiel de garder au moins deux personnes présentes pendant la maintenance ou l'exploitation.
DANGER	Lorsque l'équipement est sous tension, il est strictement interdit de déplacer, d'ouvrir ou de démonter la porte du conteneur de l'équipement électrique ou de toucher des pièces internes de l'équipement électrique avec de l'électricité. Avant d'effectuer des opérations de maintenance, vérifiez que l'équipement a été éteint en toute sécurité.
DANGER	Évitez les cordons de contact ou l'équipement en marche pendant les tempêtes.

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3.4 Electrical Safety

⚠ CAUTION	After performing AC and DC double-side disconnect operations on the energy storage system, wait at least 15 minutes before opening the front door to perform maintenance, overhaul, or other functions on the energy storage system.
NOTICE	When the energy storage system is electrically connected for a test run or other operations, use high-quality measuring equipment with a range and accuracy that meets the site requirements to ensure that the electrical parameters meet the requirements. Ensure that measuring equipment is correctly connected and used according to standards to prevent arcs and other hazards.
⚠ DANGER	Even if all external circuits are disconnected, high voltage may still exist between the positive and negative terminals of the energy storage battery pack. If measurements must be conducted under charged conditions, use dedicated instruments, and ensure that only qualified professionals perform the tests.
⚠ WARNING	If the energy storage system operates abnormally or or requires maintenance or an, shut down the equipment and confirm that the energy storage system completely stops running before proceeding with any maintenance or repair work.

⚠ CAUTION	Après avoir effectué des opérations de déconnexion double face AC et DC sur le système de stockage d'énergie, veuillez attendre au moins 15 minutes avant d'ouvrir la porte avant pour effectuer la maintenance, la révision ou d'autres fonctions sur le système de stockage d'énergie.
NOTICE	Lorsque le système de stockage d'énergie est connecté électriquement lors d'un essai ou d'autres opérations, il est nécessaire d'utiliser des équipements de mesure de haute qualité avec une portée et une précision qui répondent aux exigences du site pour s ' assurer que les paramètres électriques répondent aux exigences. Assurez-vous que l'équipement de mesure est correctement connecté et utilisé

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	conformément aux normes pour éviter les arcs électriques et autres dangers.
DANGER	Même si tous les circuits externes sont déconnectés, une haute tension existe toujours entre les bornes positive et négative de la batterie de stockage d'énergie. Si certains paramètres de mesure doivent être effectués dans des conditions chargées, le processus de test nécessite des instruments dédiés, et la mesure est effectuée par des professionnels.
WARNING	Lorsque le système de stockage d'énergie fonctionne anormalement ou qu'une révision et une maintenance sont nécessaires, arrêtez l'équipement et confirmez que le système de stockage d'énergie cesse de fonctionner avant l'entretien ou la révision.

Grounding

The following requirements apply to equipment that must be grounded:

- During installation, it must be grounded first.
- During removal, the ground wire must be the last item disconnected.
- Before operations, check the electrical connections to ensure that the equipment is reliably grounded. Operate the equipment without a grounding conductor is strictly prohibited.
- The equipment must remain permanently connected to a protective earth. Destruction of the grounding conductor is prohibited. (HyperStrong uses TN-S protective earthing system).

3.5 Operational Safety

NOTE	Read all equipment information and instructions carefully and completely before operation.
NOTICE	Before operation, please make sure that all cables are correctly connected. During operation, please pay attention to the normal operation of the equipment fan. Do not block the vent.
DANGER	Do not place small metal devices such as crank pins, screws, etc. next to connectors, slots, sockets, or circuits to prevent

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	short circuits.
CAUTION	If the equipment gets damaged, please do not repair it yourself. Contact HyperStrong to assign professional personnel to repair. Do not modify the internal equipment of the battery system privately to avoid danger.

NOTE	Lisez attentivement et complètement toutes les informations et instructions sur l'équipement avant de l'utiliser.
NOTICE	Avant l'utilisation, assurez-vous que tous les câbles sont correctement connectés. Pendant le fonctionnement, veuillez faire attention au fonctionnement normal du ventilateur de l'équipement. Ne bloquez pas d'évent.
DANGER	Ne placez pas de petits appareils métalliques tels que des manetons, des vis, etc. à côté de connecteurs, de fentes, de prises ou de circuits pour éviter les courts-circuits.
CAUTION	Si l'équipement est endommagé, veuillez ne pas le réparer vous-même. Contactez HyperStrong pour affecter du personnel professionnel à la réparation. Ne modifiez pas l'équipement interne du système de batterie à titre privé pour éviter tout danger.

3.6 Electrostatic Protection

Accumulation of static electricity can result in electric shocks, fire, explosion, failure and damage of electronic devices, and other safety hazards. The energy storage system contains circuit boards and other static-sensitive components. To mitigate the risks associated with static electricity, operators must implement static protection measures, minimize static electricity generation, accelerate the dissipation of static electricity, and neutralize static electricity.

NOTICE	The prevention methods include but are not limited to: ● During replacement, all equipment devices that have not yet been installed should be retained in bags with anti-static shielding, and temporarily removed equipment devices should be placed on foam pads with anti-static features. ● Do not touch soldering points, pins, or exposed circuits.

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EMC related cautions

The system's emission characteristics make it suitable for use in industrial areas, meaning Class A electromagnetic environment. If used in a residential environment, where Class B is typically required, this equipment may not provide adequate protection to radio-frequency communication services. User may need to implement mitigation measures, such as relocating or re-orienting the equipment.

Mains power quality should conform to the standards of a specialized industrial distribution network. For users requiring uninterrupted operation during or after power mains interruptions, it is recommended to connect the system's information storage components to an uninterruptible power supply (UPS) or a backup power source.

Some portable RF communications equipment such as high-power walkie-talkie may ddegrade the electromagnetic performance of the system. If the essential system performance is lost or impaired due to Electromagnetic disturbances, maintain a safe distance between high-power transmitters and the system, and contact customer service to address the issue.

3.7 Lifting and Transportation Safety

- When walking on top of the equipment, strictly follow the procedures for working at heights.
- Wear appropriate personal protective equipment (PPE) as required, including safety helmets, work clothing, safety shoes, and gloves.
- When using lifting machinery, adhere to all relevant safety operation guidelines. Do not engage in improper lifting practices.
- Protect the equipment during handling and storage. Prevent it from falling, tumbling, or being subjected to excessive pressure.
- Batteries transported must be securely fastened to prevent severe vibration, impact, or compression.
- Drivers and transporters must undergo appropriate training. For those handling hazardous materials, training should include emergency response procedures for fire prevention, spill containment, and emergency contact protocols. They must be familiar with the items being transported and understand the correct handling procedures.

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3.8 Installation and Wiring Safety

- During the entire mechanical and electrical installation process, strictly adhere to the relevant standards and requirements of the project location.
- Electrical wiring for the system must only be operational after inspection and verification. Ensure warning signs are prominently displayed during the process, such as "Personnel Working: Do Not Close the Gate" or "High-Voltage Danger.".
- Ensure that electrical wiring equipment perforations are sealed properly after installation. The sealing must be tight, secure, uniform in thickness, and have a neat appearance.
- Operate carefully at the installation site to prevent electric shock battery short circuit.
- Wear appropriate protective equipment during installation, including helmets, work clothes, shoes, gloves.
- Pay attention to the positive and negative direction of the battery to avoid reverse installation which can lead to battery leakage, overheating, and other issues.
- Before wiring, disconnect the power supply to eliminate the risk of electric shock.
- When wiring, ensure obstacles are avoided to prevent damage to equipment or interference with operations.

3.9 Operation and Maintenance Safety

- Schedule maintenance activities during favorable weather conditions. Avoid conducting maintenance in rain, snow, typhoons, or other adverse weather.
- Personnel must wear appropriate Personal Protective Equipment (PPE) before beginning operations, including helmets, insulated gloves, long-sleeved coveralls, and insulated shoes.
- Keep an up-to-date emergency contact list on site and ensure communication devices are operational.
- During the operation and maintenance, at least two staff members should be present on site. Confirm the maintenance tasks in writing beforehand and strictly follow the outlined operational procedures.
- After completing maintenance or overhaul, securely store the tools and restore the site to its original condition to ensure the equipment can operate as intended.
- If the ambient temperature of the battery cabin is lower than 0°C, the air conditioner will automatically switch to heating mode to preheat the cabin. In case of abnormalities, contact HyperStrong professional team for assistance.

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4 Transportation and Storage

4.1 Cautions

CAUTION

Failure to transport and store the ESS in accordance with the requirements outlined in this manual may may result in the warranty being voided.

4.2 Type of Transportation

- ESS containers can be transported by road, sea, and rail. With an integrated and user-friendly lifting design, ESS containers are easy to handle during transportation. However, air transportation is not permitted, and specific guidelines for rail transportation are not currently available.
- The ESS container can be transported domestically by truck.

NOTE: In most cases, when transported by truck, the total weight of the truck, including the ESS container, may exceed the limitations allowed by the general roadways. In such instances, an overweight permit may be required in the shipping region.

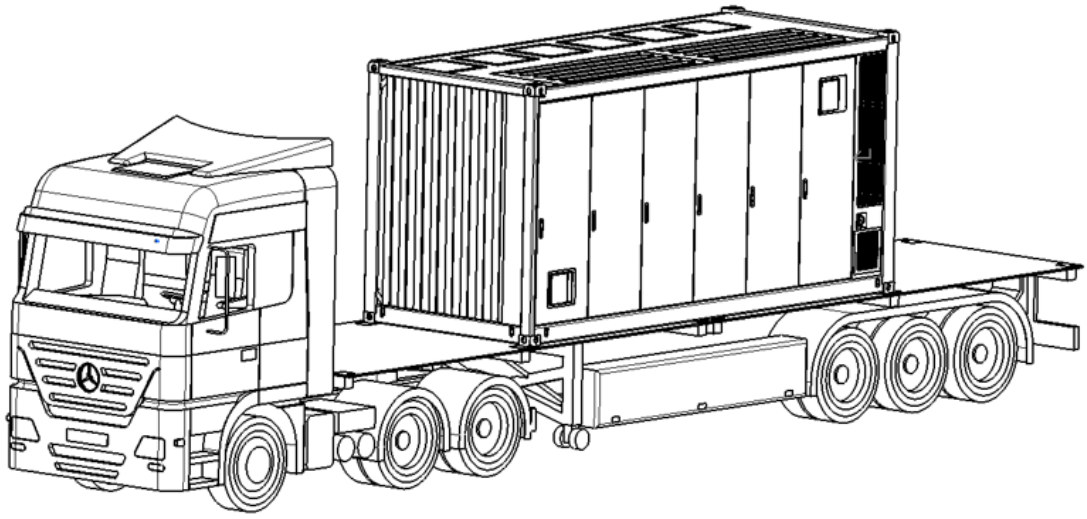


Figure 4-1 Schematic diagram of battery cabin transportation

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4.3 Transportation Requirements

The equipment in the ESS has been installed and securely fixed inside the container before shipment, allowing the container to be lifted and transported as a single unit.

- During loading, unloading and transportation, ensure compliance with the safety regulations governing ESS container operations in the country where the project is located.
- The ESS container and any machinery used in the operation should be properly maintained.
- All personnel involved in loading, unloading and bolting must be trained accordingly, especially safety training.
- The mechanical parameters of the battery prefabrication pods, including external dimensions and weight, must be accounted for throughout the loading, unloading and transportation process.
- The following conditions must be fulfilled for the transportation of ESS container:

– The doors of all cabinets in the battery container are securely locked.

– Select a suitable crane or lifting tool according to the site conditions. The selected device must have sufficient load capacity, boom length and swivel radius.

– When lifting the ESS container, it is recommended to use a single crane.

– Additional towing devices may be required if moving for slopes.

– Clear all obstacles that may exist during the movement, such as trees, cables, etc.

– The ESS container should be transported and moved under favorable weather conditions whenever possible.

– Use warning signs or tape to restrict access to the lifting and transportation area and prevent accidents.

When transporting by land, use ropes to secure the corner fittings on top of the ESS container to the vehicle. Ensure the ESS container remains stable and the inclination angle is minimized during transportation.

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4.4 Storage Requirements

- To prevent possible condensation or its bottom soaking from rainwater in the rainy season. The ESS should be stored on higher ground.
- Raise the container base are necessary to suit site conditions. The specific height should be reasonably determined based on site geological and meteorological conditions of the site.
- Store the ESS on dry, flat, and stable ground with sufficient load-bearing capacity and free of vegetation.
- Ensure the container doors and all internal equipment are securely locked.
- Storage environment temperature: -35~60℃.
- Long-term storage is not recommended because it may cause a decrease in battery capacity. Even when stored at the recommended storage temperature, irreversible capacity fading will still occur over time. The longer the storage period, the greater the capacity fade. Refer to the technical protocol for specific capacity fade rates.
- The relative humidity should be between 0~95%, without condensation.
- Protect the air inlet and outlet of the ESS to prevent rain water, sand, and dust from entering the container.
- Perform periodic inspections.

  -Check the container and the inner equipment for damage at least once per month.

  -For containers stored for more than six months, inspect the interior by opening the door to ensure there is no visible condensation. Verify that the container and internal equipment are undamaged.

  -Inspect the system after powering it on and initiating startup procedures.
- If required, request professional assistance for testing before installation.
- For BESS systems stored for more than **six months** under the above conditions, a charge and discharge cycle must be performed to bring the system SOC to **30%-40%**. The storage period begins from the date of departure from the HyperStrong factory.

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5 ESS Product

5.1 Product Introduction

The ESS utilizes lithium iron phosphate battery cells known for their high safety and long cycle life. The system is housed in an outdoor cabin, designed for modularity, easy installation, and straightforward maintenance.

A battery cell rated 3.2V/314Ah is configured into a 1P104S pack. Four packs are combined to form a rack, configured as 1P416S. Each ESS can accommodate

up to 12 sets of racks,

1 set of System Control and Distribution Cabinet (SCDC).

1 set of thermal management system (TMS).

1 set of fire suppression system (FSS).

The battery system adopts a multi-level modular design, consisting of battery cell-module-cluster-system. It connects to the SCDC via power cables.

Table 5-1 Product Parameters

Model	HSL3C7001-03343EU/US	HSL3C7001-03761 EU/US	HSL3C7001-04179 EU/US	HSL3C7001-04597 EU/US	HSL3C7001-05015 EU/US
Rated Energy (kWh)	3343.974	3761.971	4179.968	4597.964	5015.961
Rated Voltage ($V_{d.c.}$)	1331.2				
Voltage range ($V_{d.c.}$)	1123.2-1497.6				
Charge and Discharge power(kW)	1671.987 @25℃	1880.985 @25℃	2089.984 @25℃	2289.982 @25℃	2507.980 @25℃
Rated charge and discharge current(A)	1256	1413	1570	1727	1884
Dimensions (L*W*H) (mm)	6058(0,-6)*2438(0,-5)*2896(0,-5)				
Weight (kg)	31600	34200	36800	39400	42000
Applicable scope	Power grid side, new energy side, industry and commerce				

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IP rating	Pack-IP65、PDU-IP65、Container-IP55				
Operating Humidity	0~95%RH				
Operating temperature（℃）	-30~55				
Storage temperature（℃）	-35~60				
Altitude(m)	≤2000（3000 optional）				
Auxiliary power	3L/N/PE, 400/230V 50Hz (Europe) 3L/N/PE, 480/277V 60Hz (US)				
Protective class	I				
Communication interfaces	Ethernet(Optical Fiber)、RS485、Dry Contact				
Communication protocols	Modbus TCP、IEC61850、Modbus RTU				
Heat transfer	Liquid-cooling & Liquid-heating				
Coolant type and design pressure	Ethylene glycol solution@2.0Bar				
Output Short-circuit ratings	71.52kA@2ms	80.46kA@2ms	89.4kA@2ms	98.34kA@2ms	107.28kA@2ms
Fire Suppression system	Temperature sensing; Smoke sensing; Combustible gas detection; Cabin level aerosols (electric starting);				
	Ventilation system; Water fire protection				
Certifications	UL9540, UL 60730-1 Appendix H, NFPA855, NFPA68, NFPA69, UL9540A, UN38.3				
Optional	Power supply mode of fire protection system/ Explosion relief plate/Water fire protection/Communication interface of Fiber/Dehumidifier				

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Figure 5- 1 Rendering of product appearance

Salt spray resistance and seismic description

The total thickness of the BESS container paint film shall meet the C5 level, and the overall neutral salt spray resistance time must reach more than 1440h after three spraying and three drying processes, in accordance withISO12944-6 standard.

The seismic design code for BESS Container complies with GR-63-CORE, and the structural strength of container frame is suitable for seismic zone 3.

To address the diverse seismic design requirements worldwide, the most provision of design codes listed in the following table are applicable. However the precise structural and foundation designs will depend on the specific project conditions, including soil condition, rock types, ground motion parameters and relevant local laws and regulations.

Region	**Design Code**
US/Canada/Mexico	IEEE Std 693-2018
Europe	IEC 60068-3-3-2019
China Domestic	GB 50260-2013
Japan	JEAG 5003-2019
Australia/New Zealand	SAA AS 1170.4/SNZ NZS 1170.5
Chile（Reference for South America）	Nch2369-2003

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5.2 Structural Appearance and Internal Layout of ESS

Appearance of ESS container

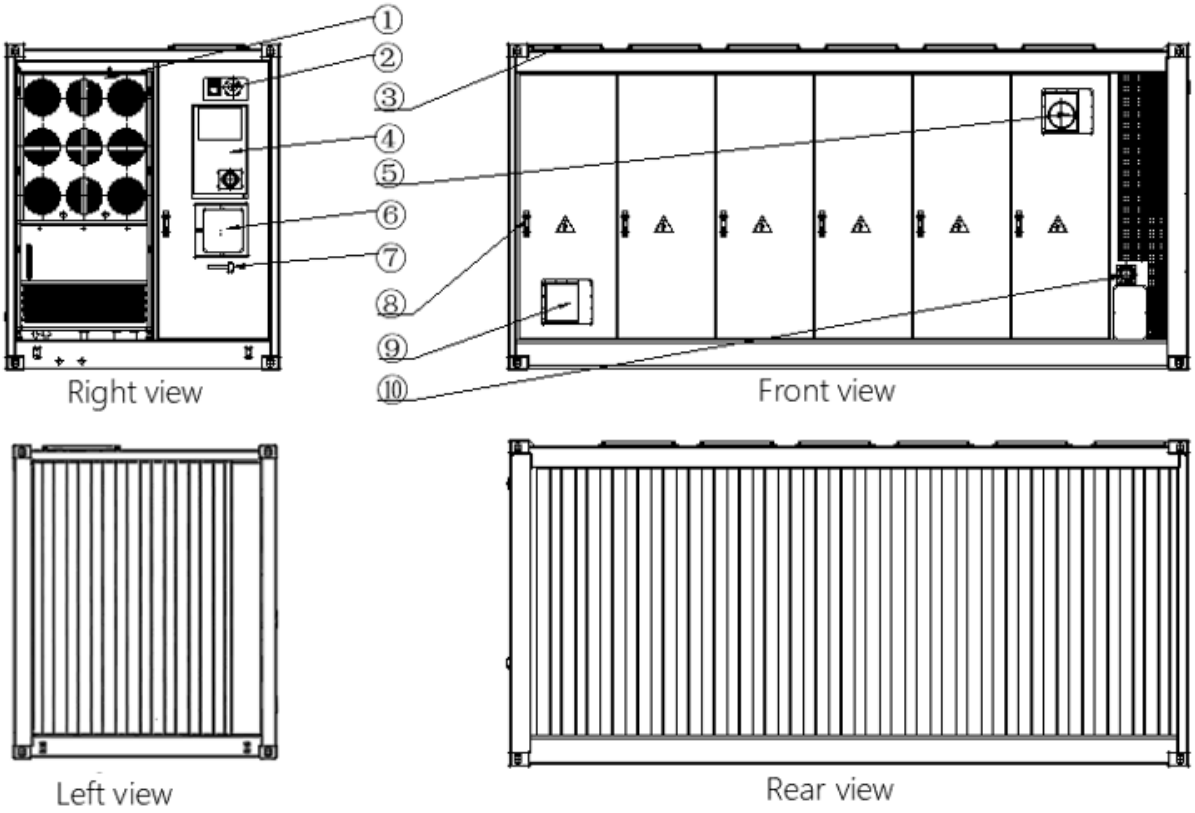


Figure 5- 2 Energy storage system cabinet appearance schematic

No.	Name
1	Cooling chamber
2	Alarm annunciator
3	Explosion vent panel (optional)
4	Air conditioning
5	Air outlet for fire suppression system
6	Emergency stop button and firefighting pushbutton Box
7	Demolition hammer
8	Door lock
9	Air inlet for fire suppression system
10	Waterborne fire protection interface

CAUTION

The nameplate contains important parameter information related to the ESS, which should be protected during transportation, installation, maintenance, overhaul and other operations. Destruction or removal is strictly prohibited.

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CAUTION

EMERGENCY STOP BUTTON: In the event of an emergency, press this button to disconnect the electrical connections inside the ESS container.

Energy Storage System Cabinet Dimensions:

The external dimensions are shown in the figure below.

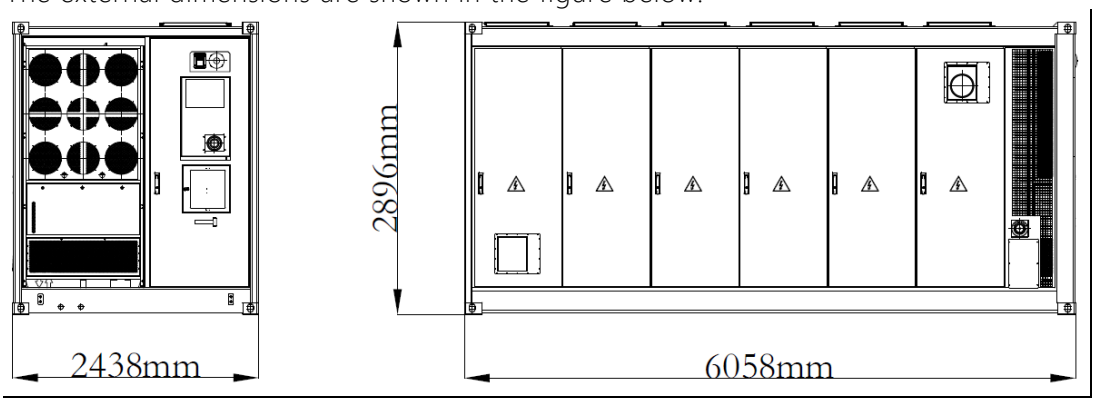


Figure 5- 3 Energy storage system cabinet dimensions

Internal Design of Energy Storage Systems:

The internal arrangement of the energy storage system is shown in the figure below.

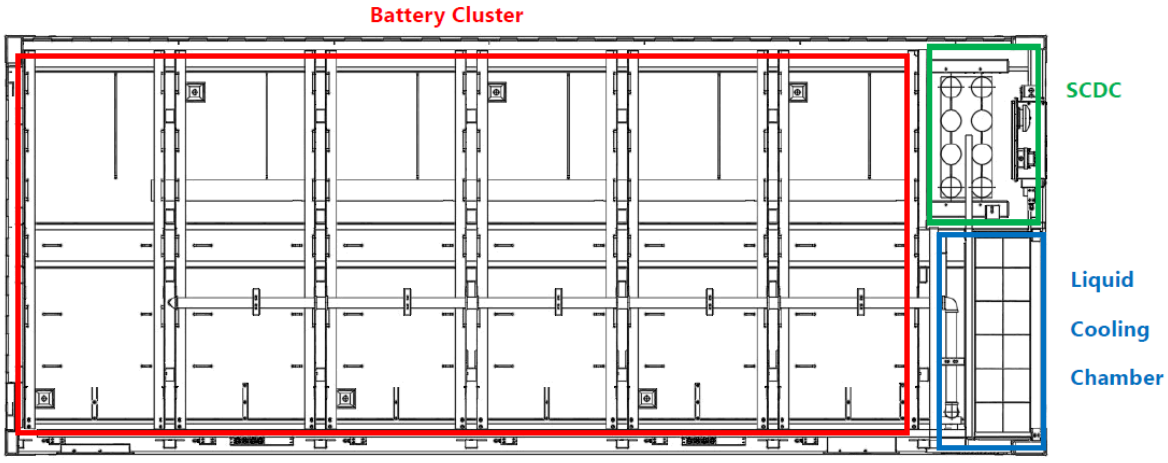


Figure 5- 4 Internal schematic of the ESS

No.	Name
1	Battery clusters
2	SCDC
3	Cooling chamber

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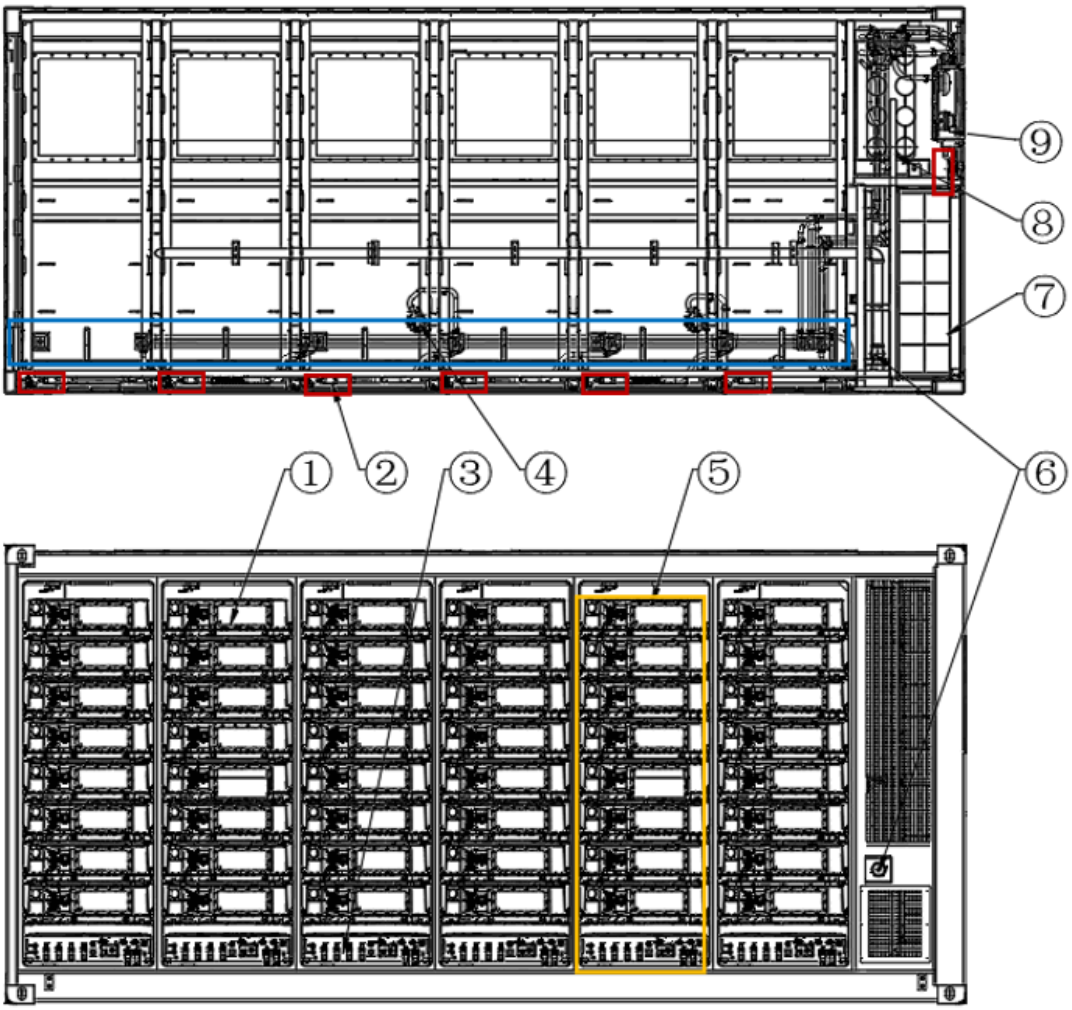


Figure 5- 5 Internal equipment composition

No.	Name
1	Pack
2	Travel switch
3	PDU
4	FSS
5	Rack
6	Water fire protection system
7	Liquid-Cooled Unit
8	System Control and Distribution Cabinet
9	HVAC
10	Temperature and humidity sensor

*The above pictures are for reference only, please refer to the actual object received.

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5.3 Primary Topology of ESS

The primary circuit of the ESS is protected by a primary main circuit, which includes an isolation switch, a surge protector, and a fuse. The isolation switch connects to 12 racks under the load, with its input connected to the DC side of the external PCS. The isolation switch enables remote closing and splitting control through the command issued by BAMU.

A schematic diagram of the product is shown in the picture below.

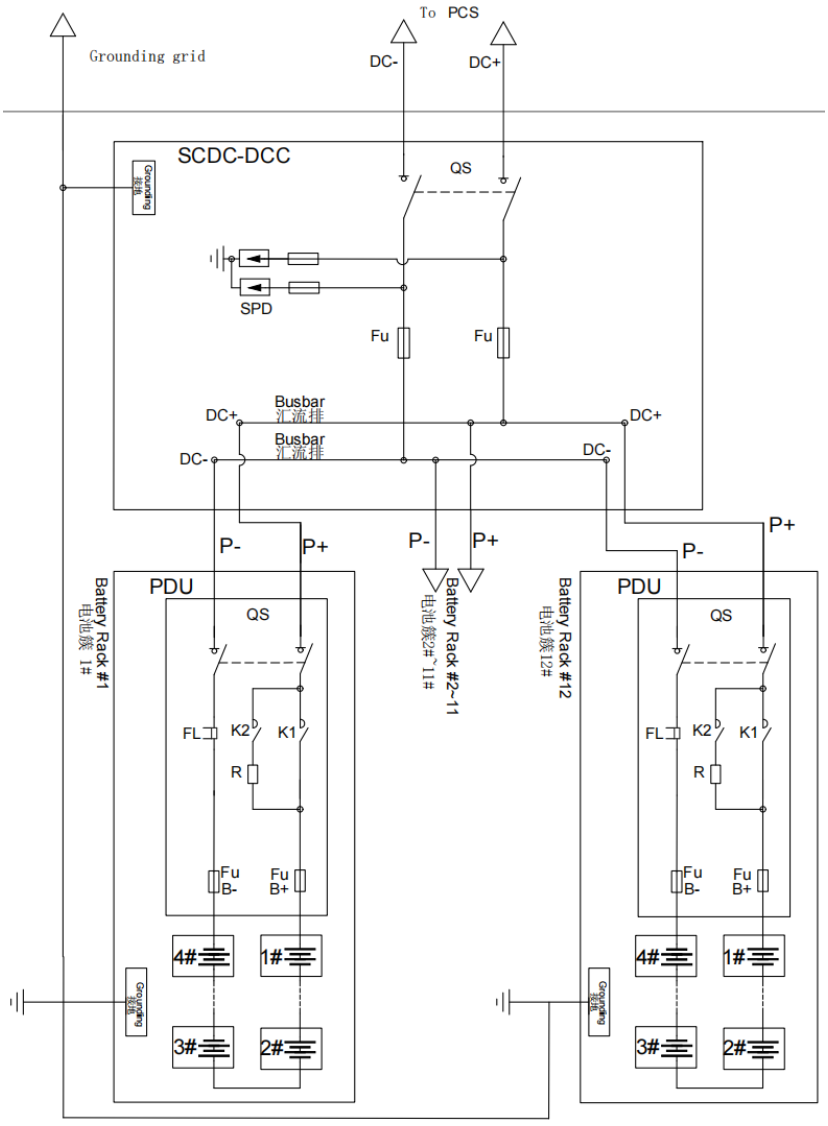


Figure 5- 6 Schematic diagram of product-12 racks

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5.4 The power supply topology of ESS

The main power supply circuit of ESS supports 400VAC/50Hz and 480VAC/60Hz power supply connections. It also includes alternative interfaces for an independent fire power supply. The alternative interfaces for independent fire power supply which supports 230VAC/50Hz and 277VAC/60Hz connections.

The power supply circuit is shown in the following figure.

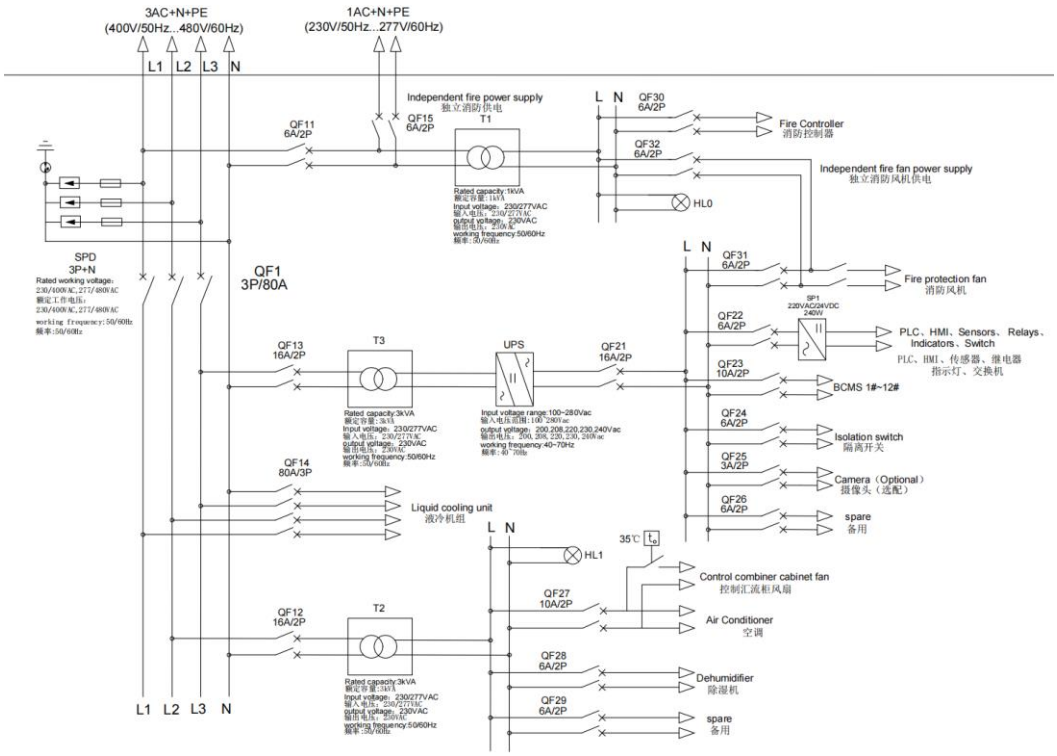


Figure 5-7 Power Supply Topology Diagram.

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5.5 Communication Topology of the ESS

The secondary topology of this product consists of BMU, BCMU, and BAMU.The communication modes between these components are:

- BMU and BCMU: Daisy chain.
- BCMUand BAMU: Ethernet.

The communication topology diagram is shown in Figure 5-7: System Communication Topology Diagram.

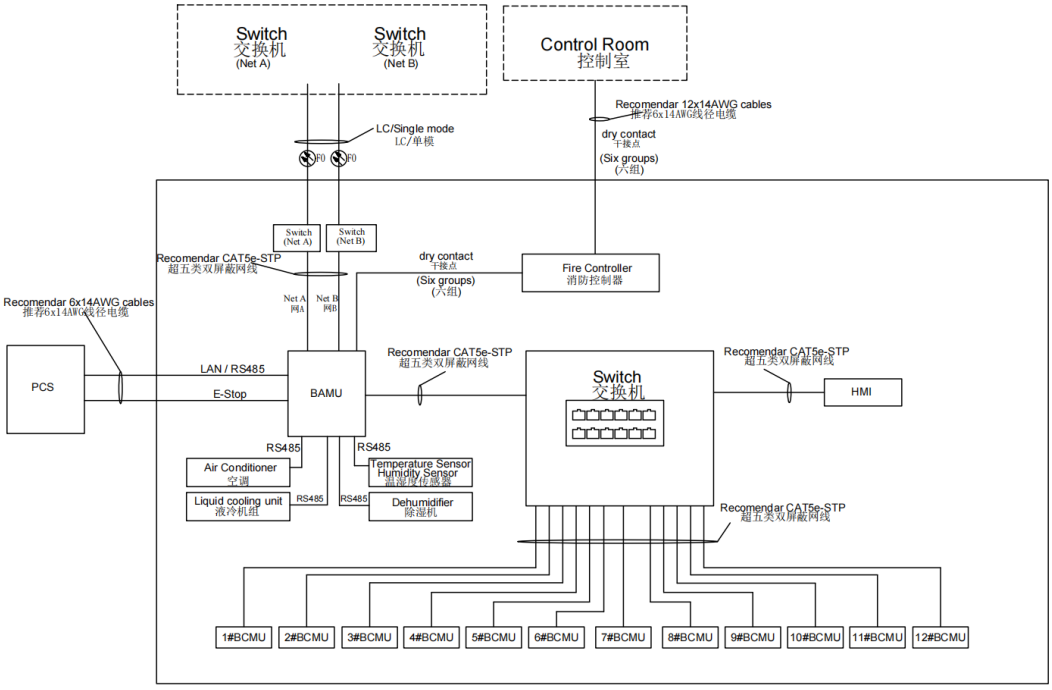


Figure 5- 8 System communication topology diagram -12 racks (for reference only)

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5.6 Rack

The rack adopts frame structure design, with the packs and PDU are fixed by bolts. Each rack contains four pack and one PDU, which are sequentially arranged in the container.

The battery rack wiring diagram is shown below: Battery Rack Writing Diagram.

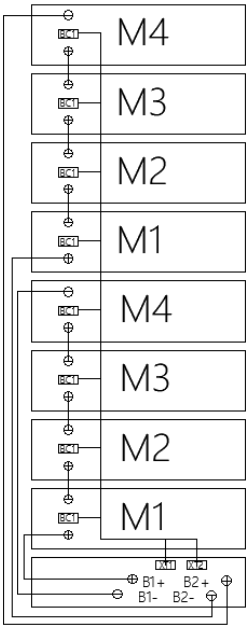


Figure 5- 9 Battery rack wiring diagram (for reference only)

Product name	Rechargeable Li-ion Battery System (Rack)
Cell Type	LFP314Ah
Cell Quantity	416
Nominal voltage	1331.2Vd.c.
Rated capacity	314Ah
Total Storage Energy	417.996kWh
Rated/Max.Charge/Discharge Power	208.998kW
Connecting type	(104S)4S
Upper limit charging voltage	1497.6Vd.c.
Final discharge voltage	1123.2Vd.c.
Nominal Current	157A
Recommend Current	157A
Over voltage category	OVC II
Operating temp.	-20°C to 55°C
Storage temperature [ºC]	-35ºC to 60ºC

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Dimension	H961.5±2, W817±2, L2198±3 (mm)
Weight	≤3010kg
Discharge cut-off voltage [V]	1123.2
Enclosure Protection (IP)	IP65 for PDU, IP65 for Battery packs
Heat transfer	Liquid-cooling & Liquid-heating
Liquid coolant type	Ethylene glycol solution
Altitude	≤2000m

5.7 Pack

The pack is designed as a battery box with a well-organized structural layout, attractive appearance, and excellent heat dissipation performance. Each 1P104S battery pack consists of eight 1P13S battery modules, collection harness, BMU, related electrical components and structural components, etc.

The battery pack is shown in Figure below Battery Pack Diagram.

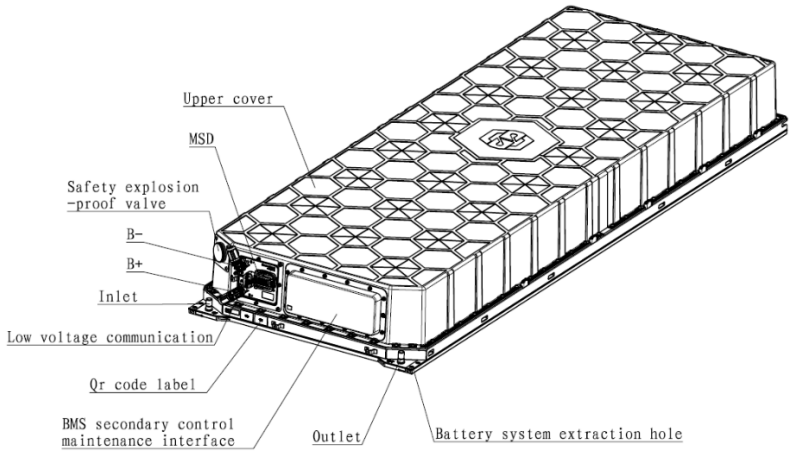


Figure 5- 10 Battery Pack Diagram

Parameter name	Parameter value
Product Name	Rechargeable Li-ion Battery Module
Cell Type	LFP314Ah
Rated energy	104.499kWh
Nominal voltage	332.8 Vdc
Upper limit charging voltage	379.6 Vdc.
Final voltage / End-of-discharge voltage	265.2 Vdc.

Recommend charging method (declared by the manufacturer)	52.249kW @25°C
Recommend discharging method (declared by the manufacturer)	52.249kW @25°C
Operating temperature	Discharge: -20 to 55°C; Charge: 0 to 55°C
Nominal mass	665±10kg
External dimensions	2190±15mm(L)*790±5mm(W)*245±5mm(H)
Cooling/heating type	Liquid-cooling & liquid-heating

*Note: Parameter values are for reference only, please refer to the actual project specifications for precise details.

5.8 Power Distribution Unit (PDU)

The Power Distribution Unit (PDU) is a control unit for distributing battery energy, containing BCMU, positive and negative contactors, fan cooling circuits, and high and low voltage connectors. All contactors are capable of receiving control commands from the BCMU, which can collect individual cell voltages and temperatures reported by BMUs within each battery pack in the cluster. After data collection and analysis, the BCMU reports the data to the BAMU. The PDU connects the high and low voltage circuits in series with each battery pack, serving as the input and output interface for the power and control circuits of the battery cluster.

The schematic diagram of PDU is shown in Figure below Schematic Diagram of PDU).

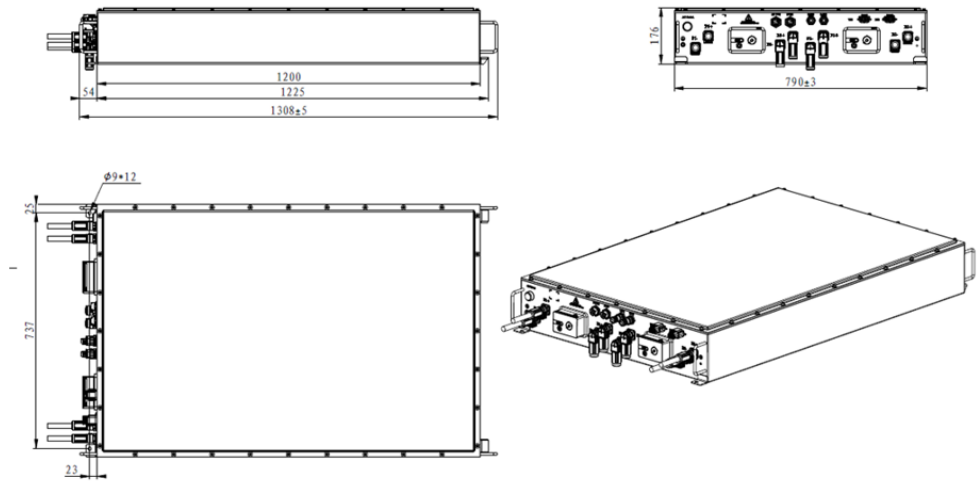


Figure 5- 11 Schematic diagram of PDU (for reference only)

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5.9 System Control and Distribution Cabinet (SCDC)

The power distribution equipment integrated with the control system and the auxiliary system has the functions of controlling the whole energy storage container and distributing power to the auxiliary system, as shown in the figure 5-11 System Control and Distribution Cabinet.

- The SCDC contains the control part and power distribution system. The power distribution part provides power supply for thermal management, BMS and other systems. The control power supply is equipped with UPS to supply power for the control loop of key equipment.
- Indoor temperature and humidity status information and fire protection status information, and upload the above data to the background.
- Try to ensure the load balance of each phase in the whole distribution system through reasonable distribution; each branch has a complete protection function, and the micro-break opening and closing status of key branches can be collected and uploaded to the background.
- Connect the BMS in the container to the switch in the SCDC, and view and control the BMS system information through the touch screen in the SCDC.
- Emergency stop under automatic or manual control can be completed in case of a fire accident or other emergency.

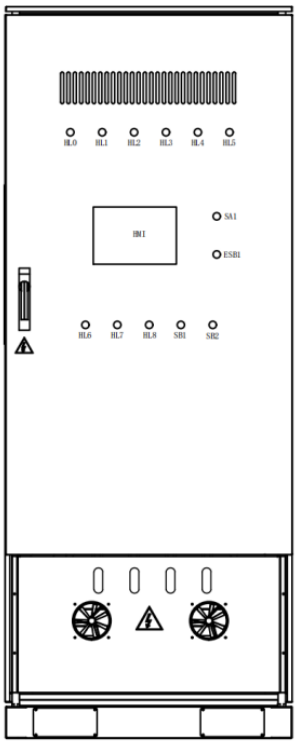


Figure 5- 7 System Control and Distribution Cabinet

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Code	Name	Indicator/ Button Color	Function Declaration
HL1	POWER		Indicates power-on status of SCDC
HL2	OPERATION		Shows the PLC is powered on and the program is in normal operation.
HL3	STOP		Indicates BAMU is in normal shutdown state
HL4	FAULT		Indicates fault warning in the BAMU in fault warning state
HL5	EMERGENCY STOP		BAMU is in an emergency stop state.
SA1	CONTROL SELECTOR LOCAL/REMOTE	/	BAMU selects local/remote control; Local state disables EMS control, and enables HMI control. The remote state does the reverse.
ESB	EMERGENCY STOP		When this button is pressed, the individual battery container unit is tripped. The SCDC will issue a emergency stop instruction to PCS and a corresponding control and protection instruction to BAMU. To restart the SCDC after an emergency stop: 1. Turn the emergency stop switch counterclockwise to release the lock. 2. Push the molded case circuit breaker to "OFF" position, then to "ON". 3. Ensure all startup condition are met, and the system will automatically begin its startup operation.
HL6	Isolation switch storage indicator light		When the self-reset state is over, the Isolation switch can be operated.
HL7	ISOLATION SWITCHON		The isolation switch is on
HL8	ISOLATION SWITCH OFF		The isolation switch is off
SA2	ISOLATION SWTICH LOCAL /REMOTE	/	Isolation switch select local/remote state; selects local/remote control; Local state disables BAMU control, and enables local panel operation. The remote state does the reverse.
SB1	ISOLATION SWTICH LOCAL OPEN		Control isolation switch opening
SB2	ISOLATION SWITCH LOCAL CLOSE		Control isolation switch closing

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HMI	Human machine interface	/	The display is used for human-computer interaction, and the user performs various operations on the ESS according to the menu window. The menu window mainly includes: rack switching, threshold setting, circuit breaker control, system configuration, IP setting, PCS information, LCS information, etc.
FAN	Radiator fan	/	Regulates the internal ambient temperature of the cabinet

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5.10 Fire Suppression System (FSS)

The fire suppression system of the ESS consists of a fire detection and alarm system, a fire extinguishing system, an active ventilation system, and a water fire protection system.

Fire detection and alarm system

Each battery **prefabricated cabin** has a set of fire detection and alarm system, which is integrated with the gas fire extinguishing system. The technical requirements of a fire detection and alarm system are as follows:

- Sensors and Detectors Combustible gas detection, smoke fire detectors, and heat detectors are installed in the battery compartment.
- Sound and Light Alarms: The system has a prefabricated sound and light alarm outside the cabin. When the gas fire extinguishing apparatus is activated, the alarm issues a warning and transmits the alarm signal to the maintenance personnel at the station or remote operation center according to the degree of fire.
- Manual and Automatic Modes: The system supports both manual and automatic mode switching. An emergency start-stop device is installed at the entrance of the emergency escape passage. Fire extinguishing can be initiated manually and false alarms can be aborted by pressing the manual abort button within 30 seconds. Afterwards, the fire alarm signal can be reset from the fire control panel, the fire controller will halt any spraying output.
- A manual extinguishant disable switch is installed at the entrance of the emergency escape passage. When maintenance is required, the system can be switched to extinguishant disable mode. （only US version）
- The electrical cabin is equipped with a dedicated fire control panel.

Aerosol Fire Extinguishing System

The aerosol fire extinguishing system consists of an aerosol (electric start) full-submergence device, sound and light alarm, emergency start-stop button, a manual/automatic switch, etc.

The aerosol fire extinguishing system has the following features:

- Two control functions including automatic, manual(Including emergency start/stop).
- Higher fire extinguishing efficiency and rapid response time.
- Fully automatic design to prevent operation errors.
- High safety, mechanical-failure free, and low maintenance costs.
- All power supply and control cables are independently arranged and galvanized steel pipes or metal channel boxes for added protection.

The control logic of the aerosol fire extinguishing system is as follows:

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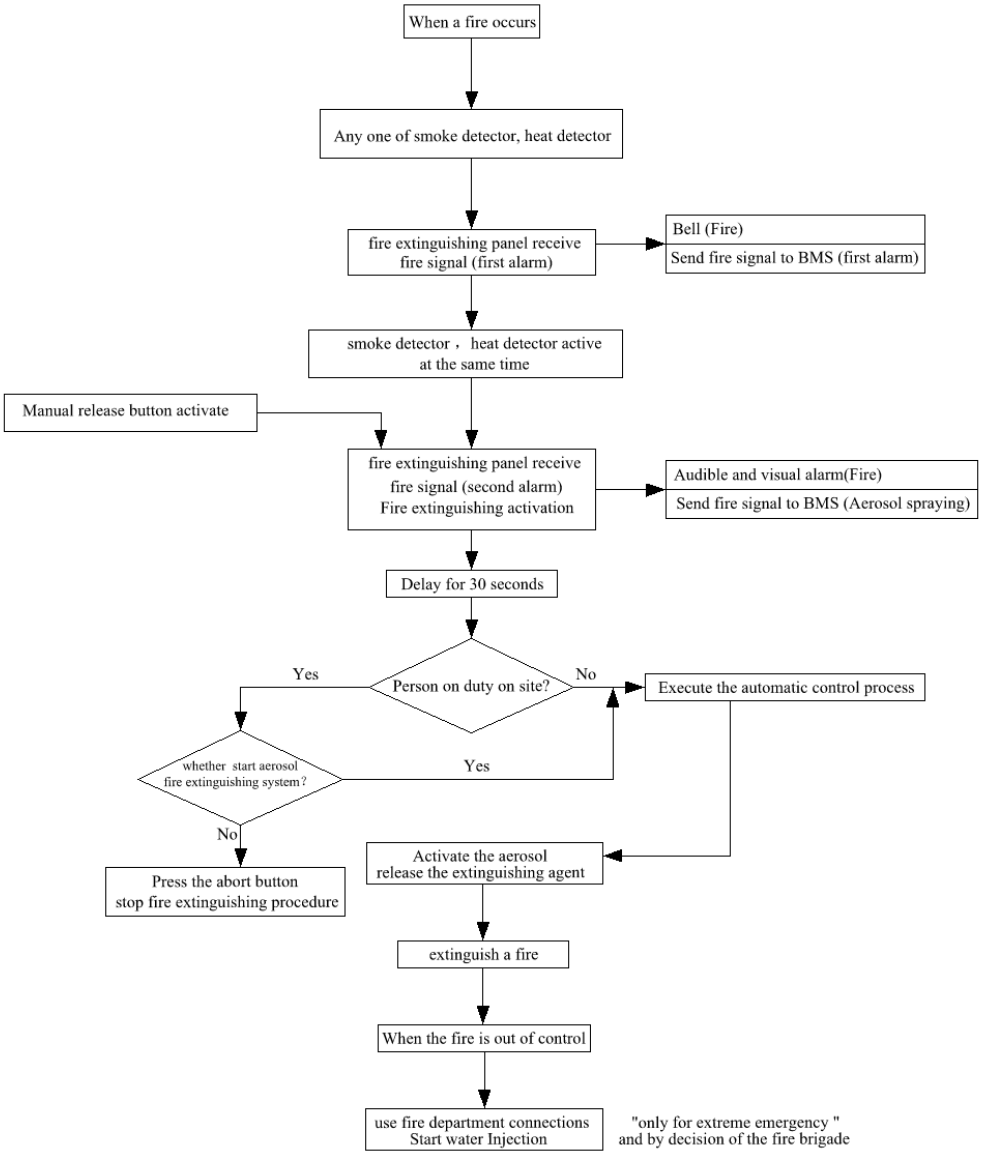


Figure 5- 13 The aerosol extinguishing system working principle flow chart

Remark:

1. The automatic aerosol fire suppression system is not intended to suppress lithium-ion battery fires but designed to suppress ancillary fires, external to the battery system.

2. Inspection, testing, and maintenance of the aerosol fire suppression system should be in accordance with NFPA 2010.

Active ventilation system

The container is additionally equipped with an active ventilation system that works in conjunction with the combustible gas monitoring module to promptly address safety hazards caused by excessive concentrations of combustible gases. The ventilation system can be manually activated with the fan start-stop button.

The active ventilation system is closed by default to maintain the container's sealing level of IP55 or above. This prevents the entry of external dust, fans, rain, etc. ensuring the safe and stable operation of the energy storage system under normal conditions. When the combustible gas detector triggers an activation signal, the energy storage system enters the protection shutdown state. At this point, the active ventilation system activates, rapidly reducing the internal pressure and the concentration of combustible gases, thereby ensuring the safety of the entire energy storage system.

The active ventilation system features high-efficiency, ventilation capability, delivering both, safety and reliability.

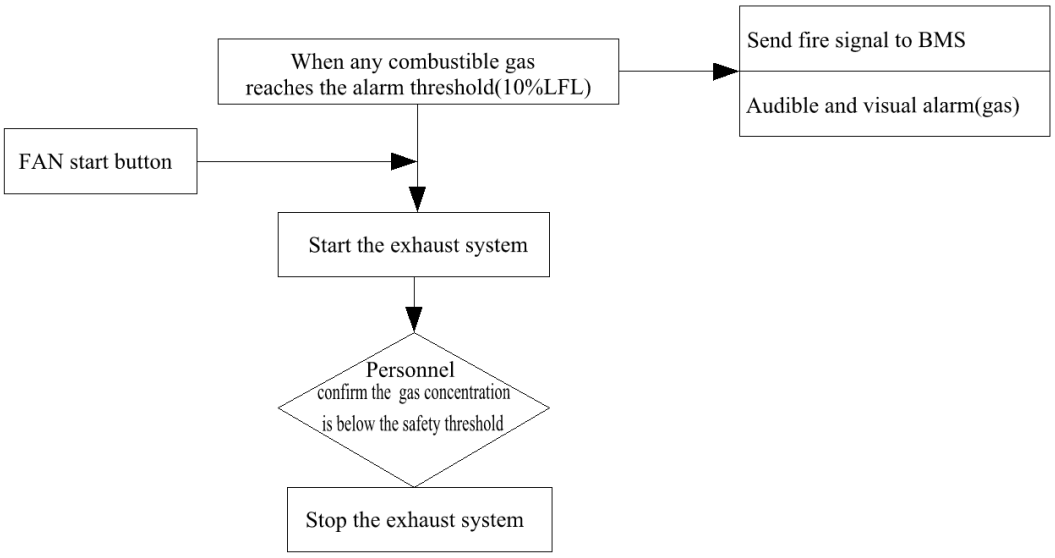


Figure 5- 14 ventilation system working principle flow chart

Remark:

1. All key components of active ventilation system related to NFPA 69, such as exhaust fan, inlet louvers, etc., will not operate at excessive temperatures.
2. Inspection, testing and maintenance recommendations for the explosion control system should be in accordance with NFPA 69.

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Explosion relief system（optional）

The explosion vent is mounted on top of the battery compartment. In the event of an explosion, when the container pressure reaches 0.1 bar, the explosion vent opens along a predetermined area to release the pressure from the container to the surrounding area.

ESS water fire protection

The rapid liquid injection system is the last line of defence for energy storage fire protection. Its primary purpose is to prevent the fire spread minimize the fire amplification and reduce damage. Each battery compartment is equipped with

One standard water fire-fighting interface.

Compartment-level piping and nozzles.

This setup enables rapid liquid injection function to ensure the safety of personnel and equipment. The system uses one 2½ "NH connector, conforming to U.S. standard hose access requirements.

Battery Fire Accident-Fire Water Access Precautions

In the event of a fire accident caused by thermal runaway, the following procedures should generally be followed:

1. The centralised control centre staff will detect the fire, call emergency services and report the fire situation and safety precautions to the person in charge of the power dispatch and operation and maintenance unit.
2. Switch off all relevant systems, except those used for auxiliary safety functions (e.g., ventilation, FACP.
3. Notify the operation, maintenance and repair personnel to arrive on site, implement safety isolation measures, and report the fire scene situation and safety precautions to the fire and rescue team commander.
4. Ensure personnel evacuate to a safe distance (NFPA 855 reference safe distance is 200ft).
5. Implement continuous remote temperature monitoring of the accident site.
6. Perform continuous air monitoring in the vicinity of the accident site.
7. In the process of extinguishing fires, respiratory protection is used throughout to avoid secondary injuries caused by toxic gases and dust particles produced by combustion.
8. After the site risk assessment, to ensure safety, firefighters may determine whether the battery compartment can be accessed for fire water source. Utilize a fire with a DN65 interface size for water injection. The water pressure should range between 0.2MPa ~ 1.2MPa, and a continuous supply of water should be applied for fire suppression, cooling and flooding.

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9. Penetrating or long ferrous nozzles should not be used in firefighting to prevent the battery packs that have not been electrically isolated avoiding risks of water penetration, short-circuits and electric shock.
10. After extinguishing the fire, do not move the damaged battery unless directed by a professional battery emergency responder, as the damaged battery module may reignite or cause arc discharge.
11. After the fire is extinguished, water should be sprayed to reduce the temperature for at least 2h to prevent re-ignition. If possible, spray water mist into the cabin for further cooling.
12. 12h after the fire extinguishing is completed, open the hatch by wearing the necessary protective equipment to ventilate and release of toxic gases Measure toxic gas concentration, and ensure the cabin temperature reaches a safe value, wearing protective equipment, enter the cabin to begin subsequent operation and analyses.

5.11 Liquid Cooling System (LCS)

The Liquid Cooling System (LCS) ensures optimal temperature control, enhancing product performance, extending battery life, reducing auxiliary power consumption, and increasing overall product value. The thermal management system is tailored to the temperature characteristics of the components in each compartment of the energy storage system. The battery room uses a coolant system for heating and cooling the batteries, while the electrical room employs forced-air cooling via air conditioners.

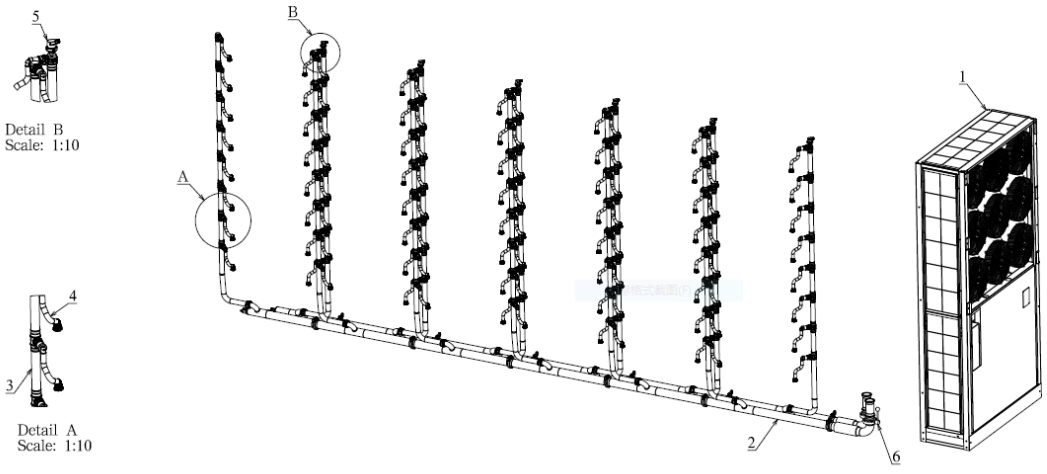


Figure 5- 8 Schematic diagram of liquid-cooled piping system for ESS

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No.	Name
1	Liquid-cooled unit
2	Primary pipe
3	Secondary pipe
4	Tertiary pipe
5	Automatic exhaust valve
6	Butterfly valve

Liquid-Cooled Unit

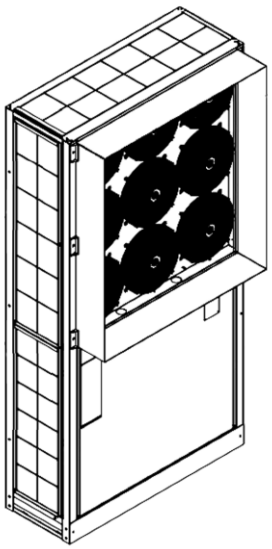


Figure 5- 16 Liquid-cooled unit

The liquid cooling unit features an integrated design, incorporating communication with the upper-level computer, alarm functionality, and other key features. This AC-powered refrigeration unit offers high reliability, simplified installation, and does not require complex debugging.

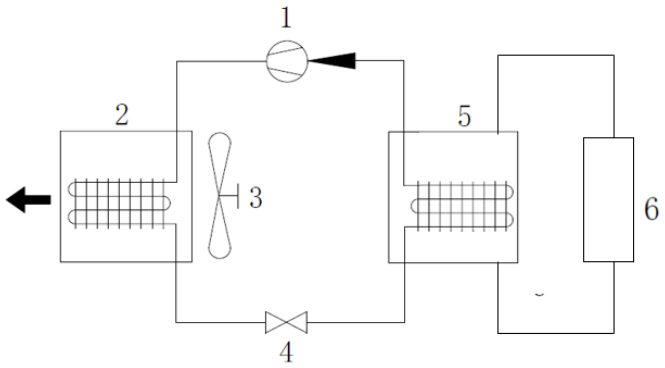
Key Characteristics of the Liquid Cooling Unit:

- The unit achieves efficient refrigeration for low-power systems through an intelligent temperature control strategy, which optimizes compressor start-up frequency and implements a load-based frequency conversion strategy.
- The unit is equipped with a heating device; the unit activates heating up when the system temperature falls below a specified threshold.
- The unit is equipped with an automatic fluid replenishment device, which replenishes fluid automatically when it is short of fluid.
- A safety valve is integrated into the system to release pressure when it exceeds safe limits, ensuring the safety of the system's pipeline.
- The system includes liquid leakage detection functionality, issuing alarms in the event of a leak to enable early identification and resolution by personnel.

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- A wind shield is added around the air outlet of the liquid cooling unit. This is to prevent the hot air blown out by the liquid cooling unit from re-entering the unit's air inlet through the installation gaps between the unit and the container, which would otherwise affect the cooling efficiency.

The operating system principle of the liquid-cooled unit is as follows



1: Compressor 2: Condenser 3: External circulation fan

4: Throttle element 5: Chiller 6: Battery coolant pipeline network

Air Conditioning System

The air conditioning equipment provides temperature control, maintaining the electrical compartment's temperature within the range of 5-35 ℃. Its key functions include.

- Cooling is activated when the ambient temperature is above 30℃, to ensure continuous and effective operation of the equipment under high-temperature conditions.
- Heating is activated when the ambient temperature is below 7℃, to prevent the startup issues caused by low temperatures.

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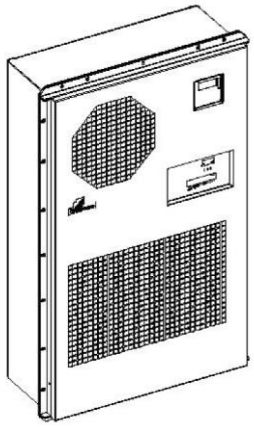
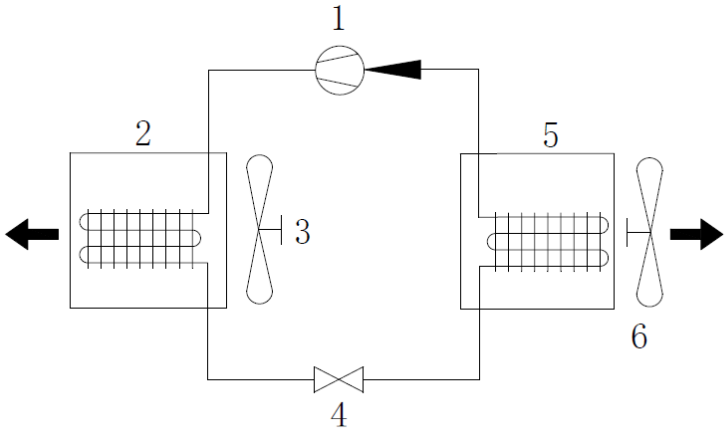


Figure 5- 17 Air conditioning system

The system principle of the air conditioning system unit is as follows.



1: Compressor 2: Condenser 3: External circulation fan

4: Throttle element 5: Evaporator 6: Internal circulation fan

Main Parameters

Technical Data	Value	Unit
Rated cooling capacity @L35 L35	2000	W
Air volume, internal circulation	650	M^3 /h
Refrigerant type	R134a	
Rated voltage	220±15%，50/60	V，Hz
IP Protection classification	IP55	
Ambient air temperature	-40~55	℃
Installation attitude	Door Mounted	
Dimensions(H*W*D)	783×483×200	mm

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Dehumidifier

The dehumidifier provides humidity control, maintaining the humidity in the battery compartment and electrical room within 80% RH.

Its key functions are as follows:

- When the ambient humidity exceeds 80% RH, it starts dehumidification to ensure that the battery system and electrical components in the electrical room can operate effectively under appropriate humidity conditions.

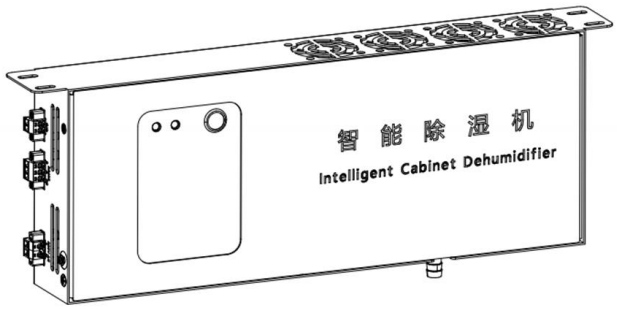


Figure 5- 9 Dehumidifier

Main Parameters

Technical Data	Value	Unit
Dehumidification capacity	1000±10%, 35℃ 85%RH	mL/D
Rated voltage	85V~265,50/60	V，Hz
IP Protection classification	IP20	
Ambient air temperature	5~55	℃
Installation attitude	Door Mounted	
Dimensions(H*W*D)	142×368×50	mm

Pipeline Network

The liquid cooling pipeline system consists of **three hierarchical components:** the primary pipe, the secondary pipe, and the tertiary pipe. The pipeline network **uniformly distributes coolant** to each battery pack, ensuring consistent heat dissipation across all battery units.

Automatic exhaust valve is installed at the top of the secondary pipe. When the pipeline pressure reaches the preset threshold, the automatic exhaust valve activates to prevent excessive pressure within the pipeline.

The liquid cooling system operates within the following pressure limits

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Maximum hydrostatic pressure: 1.2bar

Maximum working pressure: 2.5bar

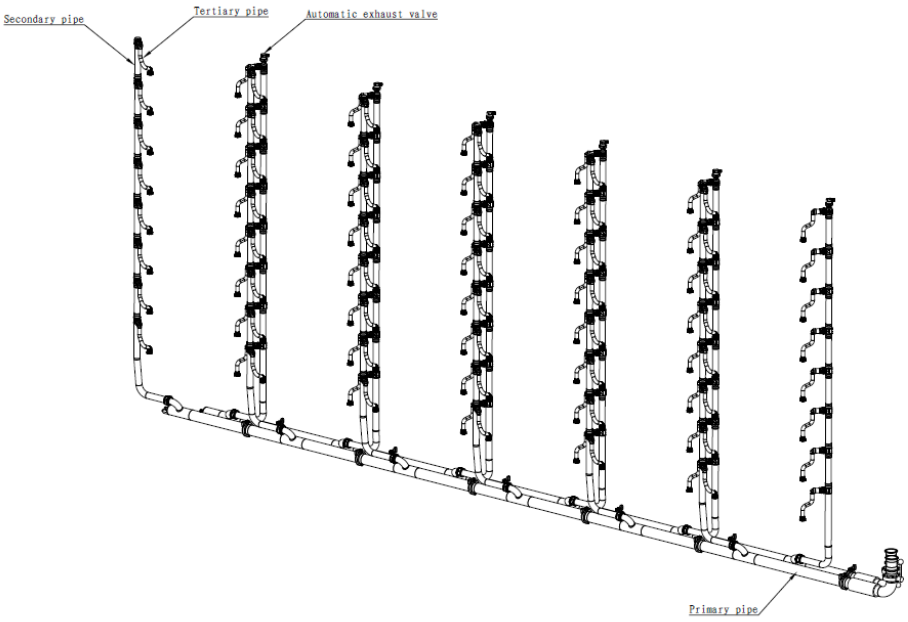


Figure 5- 19 Pipeline Network

Butterfly valve

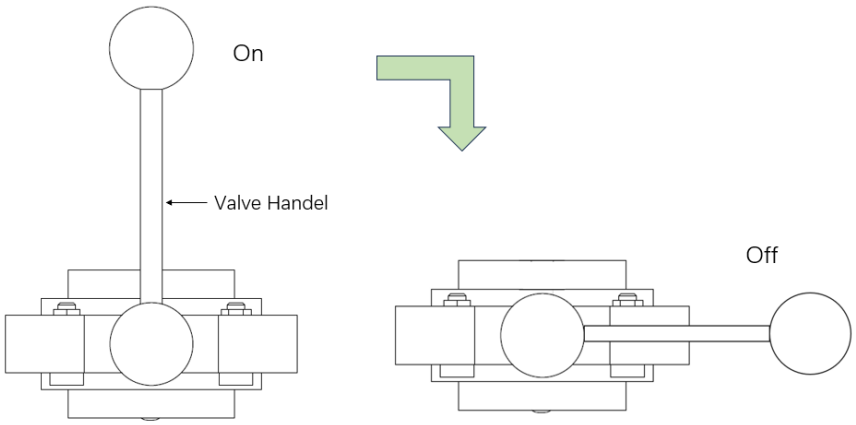


Figure 5- 20 Valves

The butterfly valves regulate the flow within the liquid cooling pipeline system, enabling convenient system maintenance.

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6 Equipment Installation

6.1 ESS Lifting

Basic information on equipment

- Overall dimensions (L*W*H): 20 feet container (high box) 6,058mm*2,438mm*2,896mm.
- Container lifting position: corner pieces, bottom lifting.
- Container Box weight: about 8.5 tons.
- Net internal load capacity: approximately 36.5 tons.
- The total weight of the battery prefabricated container (including batteries): approximately 42 tons for 12 racks. Conduct the lifting operation with a maximum load capacity of 45 tons, taking into account the variations in process and the number of modules across different versions.

Lifting Precautions

CAUTION

- When lifting and transporting, adhere strictly to all safety operation standards and norms of the country or region where the project is located.
- The Company shall not be held responsible for any personal injury or property damage resulting from the violation of relevant requirements or standards.
- Throughout the entire process of lifting the BESS, it is essential to adhere strictly to the safety operation regulations of the crane.
- Maintain a clear safety zone of **5 to 10 meters** around the operation area. Under no circumstances should personnel stand beneath the lifting arm, the load, or any moving machinery to prevent accidents.
- Suspend all lifting operations during adverse weather conditions, including but not limited to heavy rain, fog, or strong winds.

When lifting the ESS container, ensure the following requirements are met:

- Ensure the ESS container doors are locked.
- Select a crane or lifting tool suitable for site conditions. The chosen tool must meet the required load capacity, boom length, and turning radius.
- Use additional traction devices if movement on slopes or other challenging terrain is necessary.
- Clear all obstacles including trees, cables, or other potential hindrances, along the lifting path.

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- Perform lifting operations during favorable weather conditions whenever possible. Suspend operations in adverse weather, such as heavy rain, fog, or strong winds.
- Secure the site before initiating the lifting process.
- During lifting and installation operations, ensure that professional personnel are present to oversee the entire process.
- Secure warning signs or safety tape to prevent unauthorized personnel entry.
- All personnel involved in loading, unloading and bolting should receive appropriate training, particularly in safety.
- Choose the appropriate hoist based on site conditions, ensuring that the selected crane can lift the ESS safely. Recommend crane load-bearing capacity is ≥100,000 kg.
- The strength of the sling used should be able to support the weight of the ESS container.
- The length of the sling can be adjusted according to specific requirements.
- Ensure that all sling connections are safe and secure and that the sections of the sling connected to the corner pieces are of equal length.
- Adjust the length of the sling appropriately based on the actual requirements of the site.
- The entire lifting process must ensure that the battery cabin is smooth and does not deflect.
- Utilize the four telescopic posts of the ESS for lifting.
- Implement all necessary auxiliary measures to guarantee the safe and smooth lifting of the battery prefabrication module.
- When lifting the battery cabin, the crane must be operated strictly in accordance with the crane's safety regulations.
- Standing within 5~10m of the operation area is strictly prohibited. It is strictly forbidden for people to stand beneath the lifting arm or moving machine to avoid accidents.
- Cease lifting operations in adverse weather conditions, such as heavy rain, fog, or strong winds.
- The following figure illustrates the crane operation while lifting the ESS container. The inner dotted circle in the figure indicates the crane's working area. Anyone should not stand inside of the red circle during crane operation.

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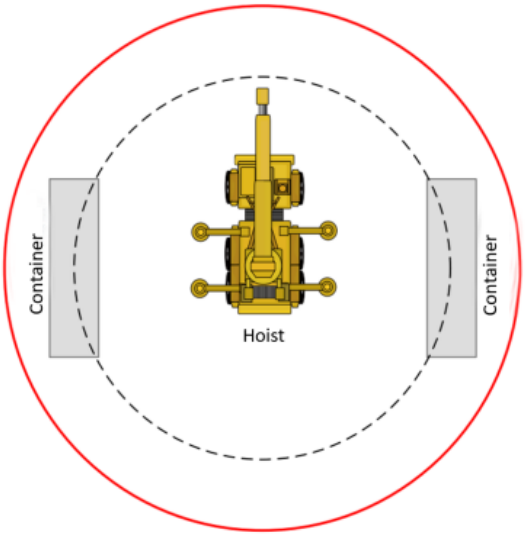


Figure 6- 1 Schematic Diagram of the Crane Operation

Container Lifting Process

During the lifting of the ESS, all operational aspects shall be carried out in accordance with the following requirements:

- The ESS must be lifted vertically, without dragging on the ground, the roof of the lower box, or other surface. Dragging or pushing the ESS on any surface is strictly prohibited.
- After lifting the ESS to a height of 300 mm from the support surface, suspend it, inspect the connection between the spreader and the ESS, and proceed with lifting only after confirming the connection is secure.
- **When placing the ESS, it must be gently lowered and landed smoothly. It is strictly prohibited to position the ESS outside of vertical alignment by flinging the spreader**.
- The installation site for the ESS must be stable, level, well-drained, and free of obstacles or protrusions.

Description of Specific Lifting Program for Containers

Special care should be taken to protect the sides of the container to prevent scratches. Refer to the following figure for lifting guidance.

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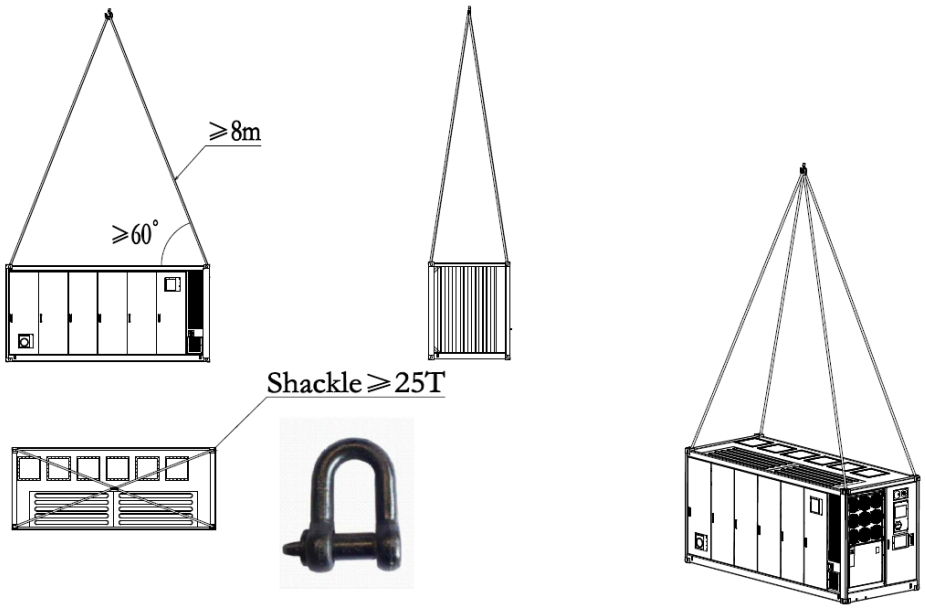


Figure 6- 2 Container support and fixing point locations

*The pictures above are for reference only. Please refer to the actual object received for details.

Part Name	Parameter
Sling	sling ≥ 8 meters, load > 25T;

Bottom fixing

Welding fixation method.

Weld and secure the container bottom to the foundation. After welding, apply anti-corrosion treatment and seal all around.

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6.2 Mechanical Installation

CAUTION

During the entire process of machinery installation, the relevant standards and requirements of the project site must be strictly followed.

Pre-Installation Checks

- Inspection of deliverables

Check the completeness of the deliverables against the accompanying packing list.

- Inspection equipment

Verify that the received cabinet corresponds to the ordered model.

Examine the ESS container and internal equipment to ensure no damage.

CAUTION

Only a complete and undamaged ESS container should be installed!
Please ensure the following before installation:

- The unit itself is intact and undamaged.
- The equipment inside the ESS is intact and undamaged.

If any issues or questions arise, promptly contact the transporter or our company.

Installation Environmental Requirements

Site Selection Requirements

When selecting the installation site, due consideration should be given to its characteristics, such as climate, and geological conditions (e.g., stress wave emission, underground water level, etc.).

- The surrounding environment should be dry and well-ventilated.
- Confirm the absence of trees around the installation site to prevent obstruction of the system's doors or air inlets by tree branches or leaves carried by high winds.
- The site should be far from areas where toxic and hazardous gases are concentrated, as well as from flammable, explosive and corrosive substances.
- The equipment should be installed far from residential areas to avoid noise.

Foundation requirements

CAUTION

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The overall weight of the ESS is significant, and the conditions of the installation site (primarily geological constraints, environmental factors, and climatic conditions, etc.) should be thoroughly examined before the constructing the foundation. Only after completing this assessment can the design and construction of the foundation begin.

An improper foundation construction plan can lead to difficulties or complications in the placement of the ESS container, the opening and closing of its doors, and its subsequent operation. Therefore, the foundation of the ESS container must be designed and constructed in accordance with specific standards beforehand to meet the requirements of mechanical support, cable routing, and later maintenance and overhaul.

When constructing the foundation, the following requirements must be met:

- The soil at the installation site must have a certain degree of compactness. It is recommended that the relative compactness of the soil at the installation site be ≥ 98%. If the soil is loose, appropriate measures should be taken to ensure the stability of the foundation.
- The bottom of the pit where the foundation is to be constructed must be compacted and filled to provide adequate and effective support for the cabinet.
- The foundation should be constructed according to the provided foundation plan or the one confirmed by HyperStrong. The flatness tolerance of the foundation should be within ±3mm. Specifically, the height difference between the lowest point and the highest point of the upper surface of the foundation must not exceed 6 mm, as shown in the following figure.

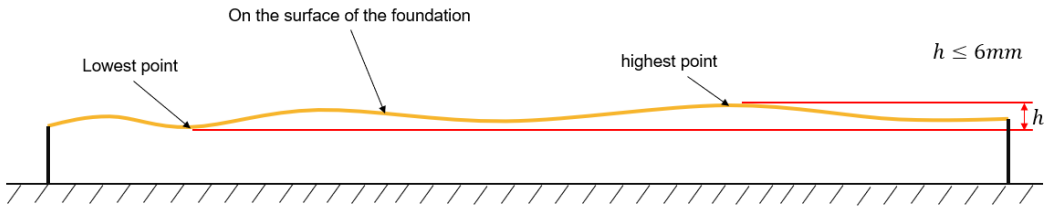


Figure 6- 3 Foundation flatness requirement

- Raise the energy storage container to prevent rainwater from eroding the base and interior of the equipment.
- Ensure that the cross-sectional area and height of the foundation meet the specified requirements.
- Drainage measures must be implemented in accordance with the local geological conditions.
- Construct a concrete foundation with sufficient cross-sectional area and height. The responsible engineering entity will determine the foundation's size based on the geological characteristics of the site.
- Cable routing should be considered during foundation construction.

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CAUTION

Silt excavated during foundation construction should be removed immediately to avoid affecting the subsequent lifting of ESS containers.

- The maintenance platform is constructed around the foundation to facilitate later maintenance.
- Based on the location and size of the upper cable entrances and exits of the energy storage containers, reserve enough space for the cable troughs on the AC/DC side and pre-embed the cable conduits during foundation construction.
- Determine the specification and number of injection tubes according to the cable type and the number of inlet and outlet lines.
- A drainage system should be constructed to prevent waterlogging at the bottom of the energy storage container or inside the equipment during the rainy season or heavy precipitation.
- Temporarily sealed both ends of all pre-embedded pipes to prevent contaminants from entering as this could complicate wiring later.
- After connecting all cables, cable entrances, exits, and joints are sealed with fire-resistant mud or other suitable material to prevent rodent entry.
- Pre-bury grounding units according to the relevant standards of the country or region where the project is located.

Restricted locations

A pre-assembled battery system shall not be installed:

a) in restricted locations as defined for switchboards in AS/NZS 3000.

b) within 600mm of any exit.

c) within 600mm of any vertical side of a window or building ventilation that ventilates a habitable room.

d) within 600mm of any hot water unit, air conditioning unit or any other appliance not associated with the pre-assembled battery system.

e) within 900mm below any of the items included (b), (c) and (d).

f) in ceiling spaces.

g) in wall cavities.

h) on roofs, except where specifically deemed suitable.

i) under stairways.

j) under access walkways.

k) in any evacuation route or escape route.

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Installation Space Requirements

To ensure optimal air intake for the air inlet, sufficient space should be allocated around the ESS container installation location. The minimum space requirement is illustrated in the figure below.

- The recommended distance between the long sides of containers is 3500mm as shown in the figure, with a minimum limit of 3000mm (the installation base shall not exceed the outline of the container).
- The recommended distance between the short sides of two containers is 150mm, with a minimum limit of 100mm.

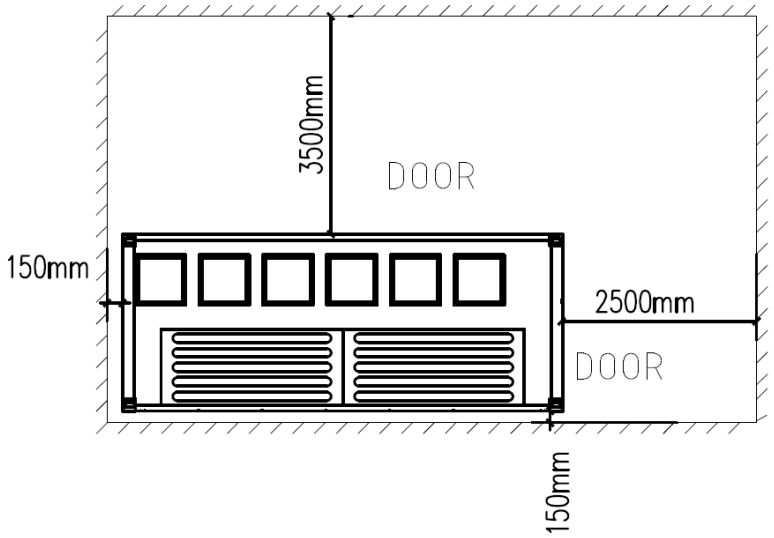


Figure 6- 4 Individual container installation

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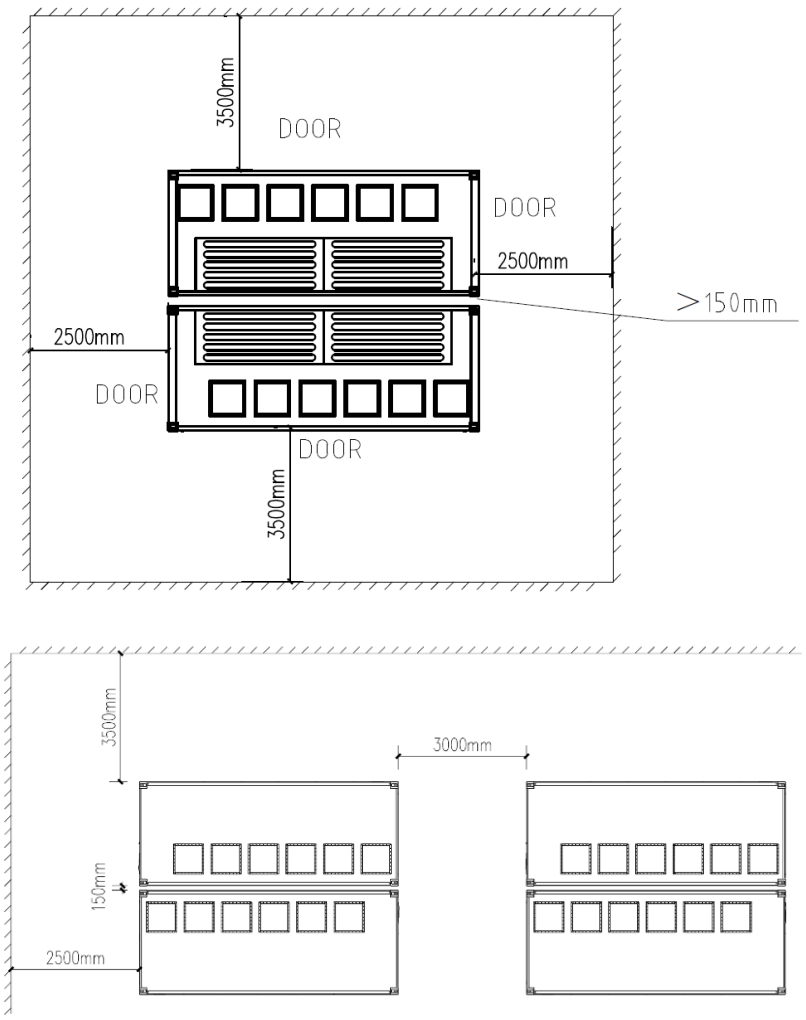


Figure 6- 5 Multi container installation

*The above figures are for reference only, please refer to the actual object received.

Pack Installation

The pack and its mounting location are shown in Figure 6-6, which is installed inside the container. A 3000mm maintenance space should be reserved in front of the installation area.

1.Align the battery pack with the guide rail.

2.Push it to the limit position.

3. Secure the battery pack onto the mounting bracket using M10 bolts to lock.

The pack fixing method is shown in Figure 6-7, which utilizes two hexagon socket head cap screws (M10 * 25mm) and hexagonal nuts to lock the pack directly.

The mounting surface of the ESS pack serves as the ground point. The contact surface between the nut, gasket at the bottom of the container pack bracket, and the pack and bolt act as the ground conductor.

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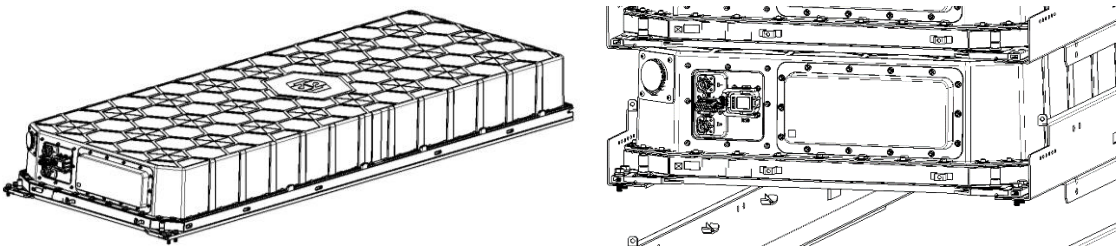


Figure 6- 6 ESS pack and mounting location

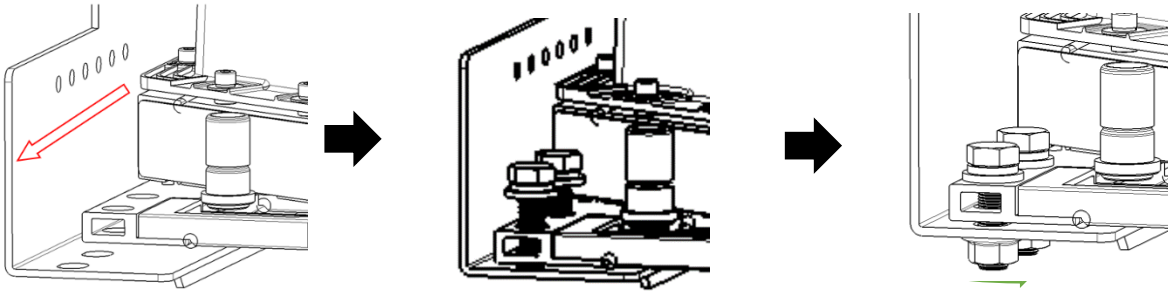


Figure 6- 7 Pack installation

System Control and Distribution Cabinet (SCDC) Grounding

The SCDC is installed in the electrical compartment inside the container. During installation, 800mm of maintenance space is required at the front, and 10mm of installation space can be reserved between the side and the sealing strip. The bottom mounting holes of the SCDC are shown in Figure 6-9. They are secured with hexagonal head screws (M10*25mm), flat washers M10, spring washers (M10) and nuts (M10).

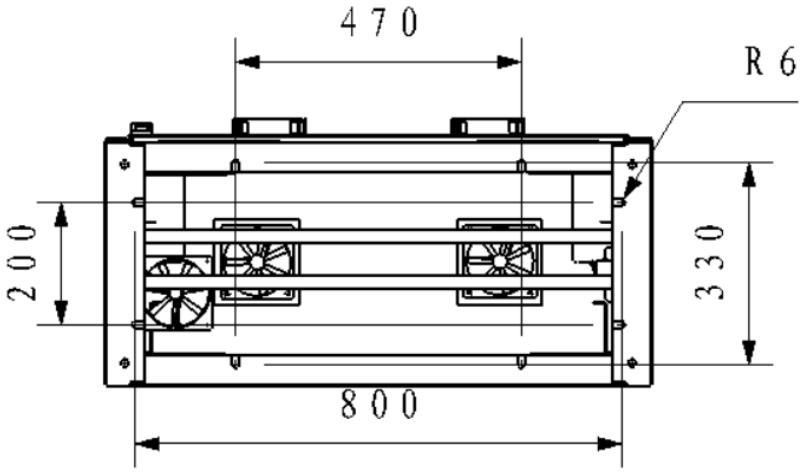


Figure 6- 8 SCDC mounting hole diagram

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The SCDC is equipped with two grounding columns located at the bottom of the cabinet, see Figure6-10. The grounding columns are M10 welded bolts with the height of 20mm. Grounding wire must be configured based on project requirements and secured using M10 flat washers, M10nuts, and M10spring washers with an installation torque of 20N.m ± 1N.m.

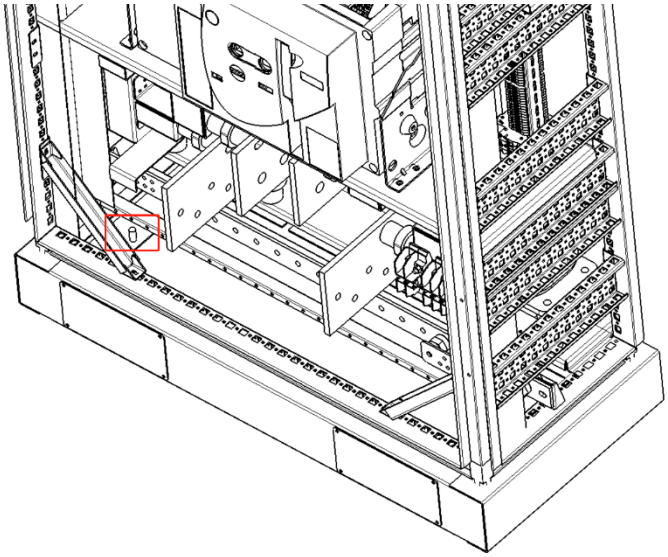


Figure 6- 9 Control bus cabinet grounding diagram

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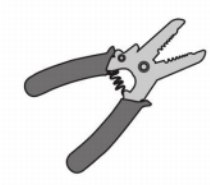
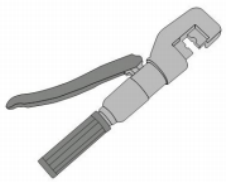
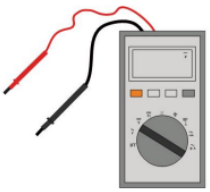
7 Product Electrical Installation

7.1 Setup Tools Preparation

Personal protective devices：

		
Safety gloves	Safety shoes	Safety helmet

Tool preparation：

		
Screwdriver	Wire stripper	Hydraulic tong
		
Heat gun	Multimeter	M10 socket wrench

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7.2 Opening Doors of Boxes and Equipment Cabinets

Step 1: Open the door of the battery cabin's electrical room.

Step 2: Remove the sealing plate bolts.

Step 3: Unplug the fan from the plug.

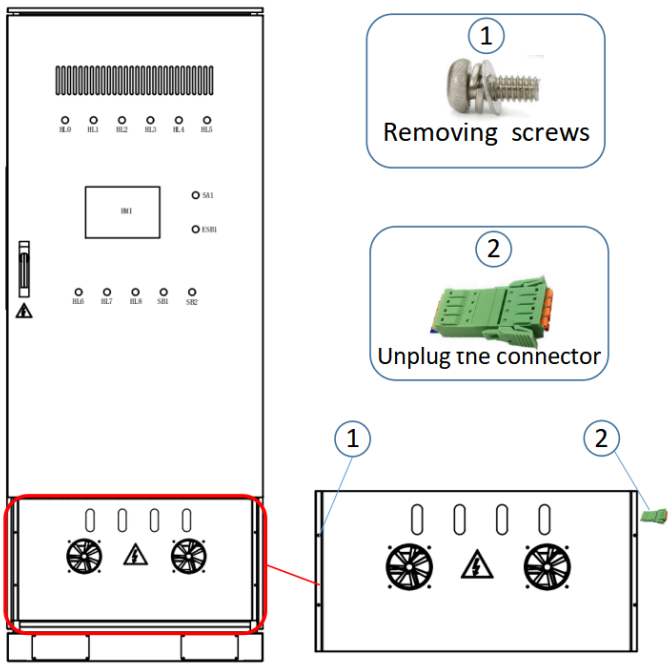


Figure 7- 1 Boxes and equipment cabinets door diagram

Key Illustration

The key box is located on the inside of the cabinet door in the electrical room (as shown in the figure. The key box contains the following items:

- Cabinet door key
- Cabinet door key of water-cooled unit
- Extinguishant control box door key
- Power control box door key
-
- Emergency release button key
- (U.S. version only): Extinguishant disablement switch)

For complete details, refer to the actual keys in the key.

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Schematic diagram of the key box

7.3 Cable Harness Preparation

The cables selected must meet the following conditions:

(1) Possess adequate current-carrying capacity. Factors affecting the current-carrying capacity of the conductor include, but not limited to, environmental conditions, conductor insulation material type, cable laying method, cable material, and cross-sectional area.

(2) The cable diameter must be selected in accordance with the maximum current-carrying capacity, and there must be a margin for the length.

(3) All DC input cables shall have the same specifications and materials.

(4) The specifications and materials of three-phase AC output cables shall be consistent.

(5) Flame-retardant cables must be selected.

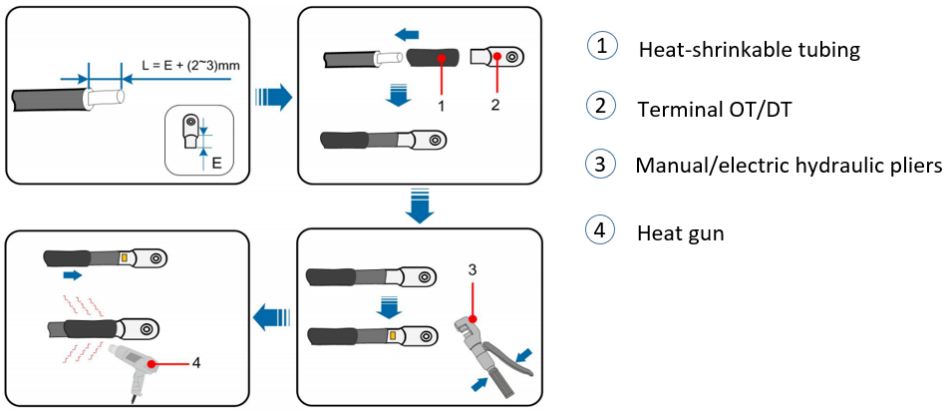


Figure 7- 2 Crimp OT/DT terminals

Note: Armored cables need to be considered with a single-ended grounding of the shield, which can be on either the ESS side or the PCS side.

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7.4 DC Cable Connections

Each rack uses 1AWG DC power cable to converge to the DC busbar, the DC lug is shown in the picture below.

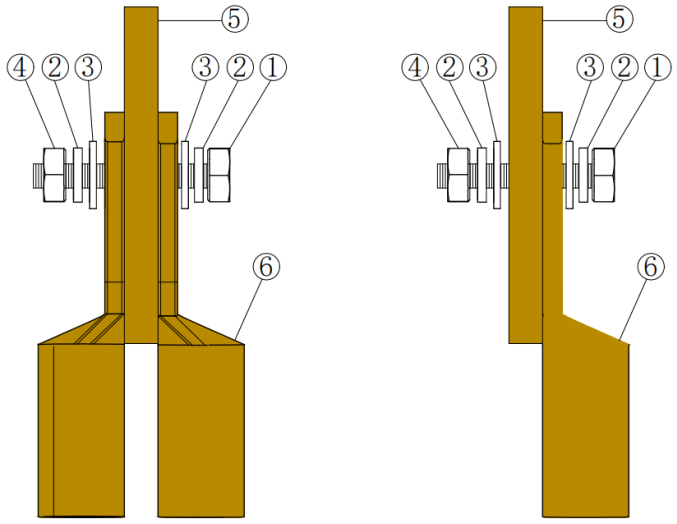


Figure 7- 3 DC connection terminal

No.	Name	No.	Name
1	Bolt	4	Nut
2	Spring washer	5	Copper row
3	Flat washer	6	Copper tip

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7.5 Cable Access Port

The cable inlet of the ESS container is located at the bottom of the SCDC, and the external access cables are uniformly fed from the bottom. The inlet is sealed by a tower-shaped sealing ring, which needs to be punctured during threading. After wiring, the inlet should be tightly sealed with fireproof mud or other suitable materials.

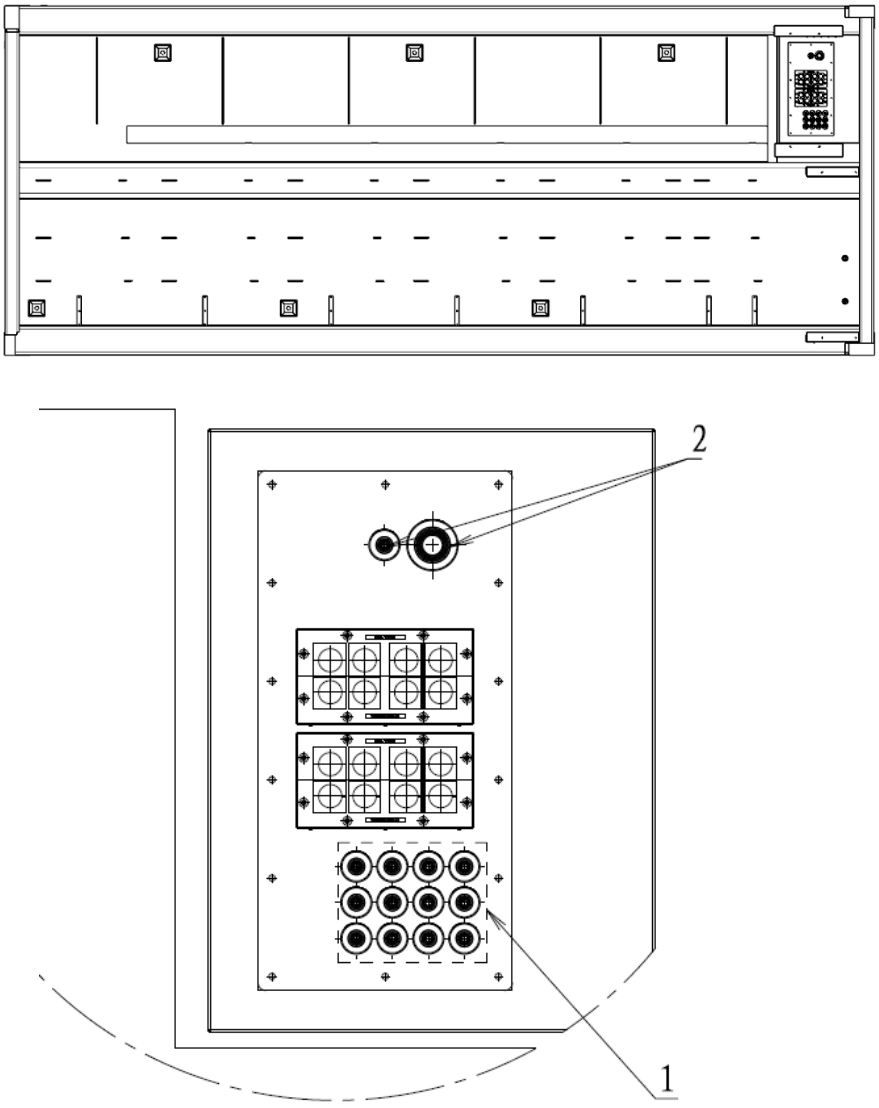


Figure 7- 4 Cable access port

No.	Connection Types	Compatible Wire Diameter(mm)	Terminal Height(mm)
1	Communication	3-12	2000
2	Auxiliary power supply	3-12/22-36	600
Other	Power	29-31	680

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7.6 Container Grounding

There are two methods of grounding: welding with grounding flat steel or fixing with a grounding cable.

- Grounding flat steel: Remove the protective tape from the earthing point and weld a 50 mm x 100 mm hot-dip galvanized flat steel to the earthing point. After completion, the entire fixing surface must be sprayed.

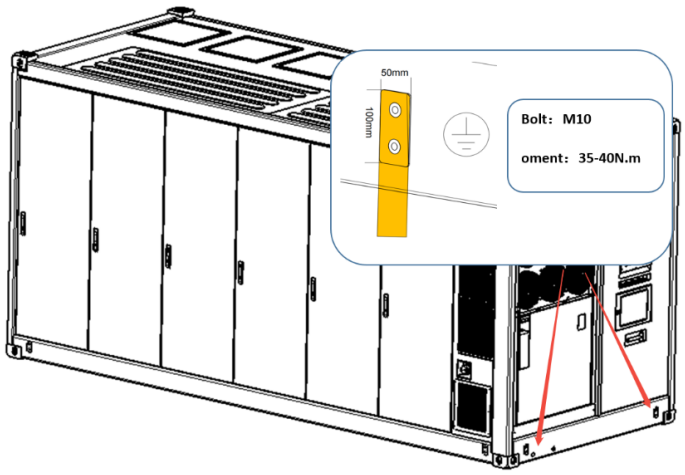


Figure 7- 5 Busbar grounding

- Grounding cable: Use a 70mm^2 ~ 95mm^2 grounding cable to reliably connect the two grounding points to the ESS grounding point. (Tape is attached to the factory grounding point, which needs to be removed before wiring.).

Crimp the terminals and secured the cables using M10 bolts.

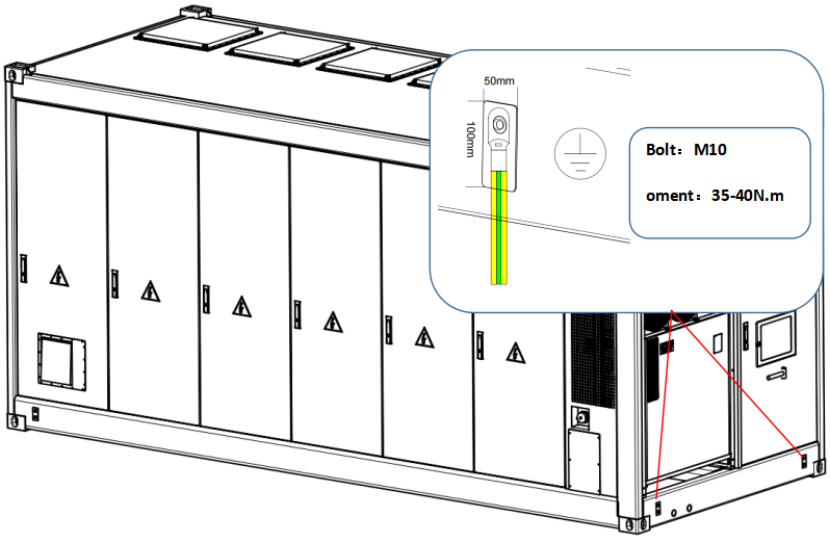
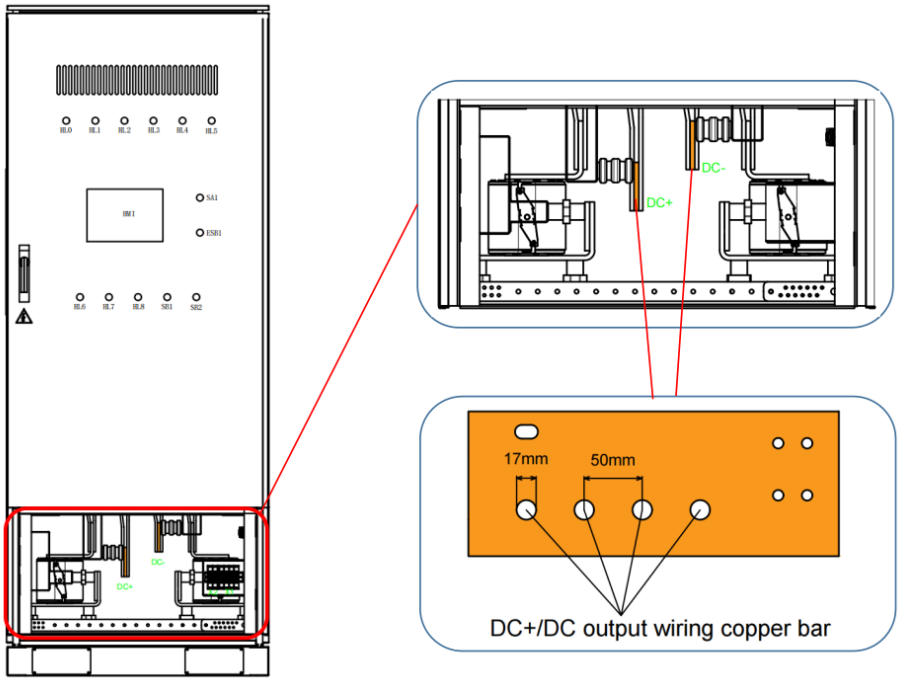


Figure 7- 6 Cable grounding

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7.7 DC Input Connection

The Battery Prefabrication Pod direct access port is shown in the figure below.



Suggest selecting 240mm ^{2} x6 for single pole DC cables.

Figure 7- 7 Access port

*The above figures are for reference only, please refer to the actual object received.

Operating procedure:

Step 1: Lead the cable into the inlet hole and into the wiring area of the copper row, ensuring the polarity of the cable is marked.

Step 2: Use wire strippers to remove the protective layer of the cable, exposing the copper core.

Step 3: Crimp the OT terminals.

Step 4: Secure the OT terminals to the wiring holes with M16 bolts, with a tightening torque is 119~140 N-m.

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7.8 AC Input Connection

The ESS AC power input connector is shown in the figure below.

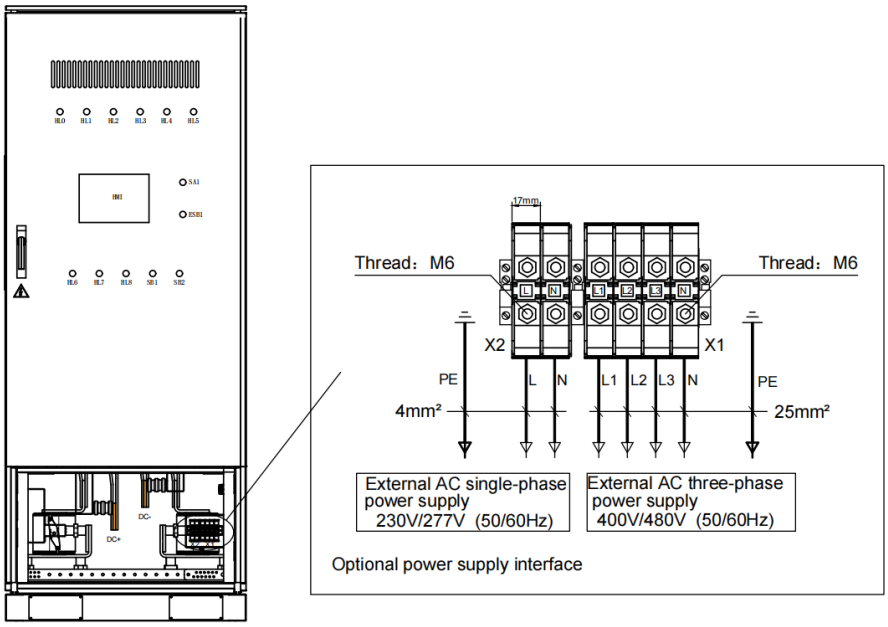


Figure 7- 8 AC Power Input Connector

*The above figures are for reference only, please refer to the actual object received.

No.	Name	Model	Picture
1	Wiring terminals	RBO 6 (3075896)	
2	Terminal side panel	D-RSC 6 (3213098)	
3	Cover plate	AP RSC-T (3059139)	

Operating procedure:

Step 1: Insert the cable into the inlet hole and enter the CT1 terminal on the right wall of the SCDC and mark the phase sequence of the cable.

Step 2: Use wire strippers to remove the protective layer of the cable, exposing the copper core.

Step 3: Crimp the OT terminals.

Step 4: Remove the M6 nut supplied with the TC1 terminal and fix the OT terminal to the wiring hole with a tightening torque of 20~25 N-m.".

Note: **Ensure wiring is performed in accordance with the labeled phase sequence**.

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7.9 Container External Communication Interfaces

Control Room and PCS Communication Interfaces

The interface between Battery Container and Station Level Fire Protection Host is shown in the figure below.

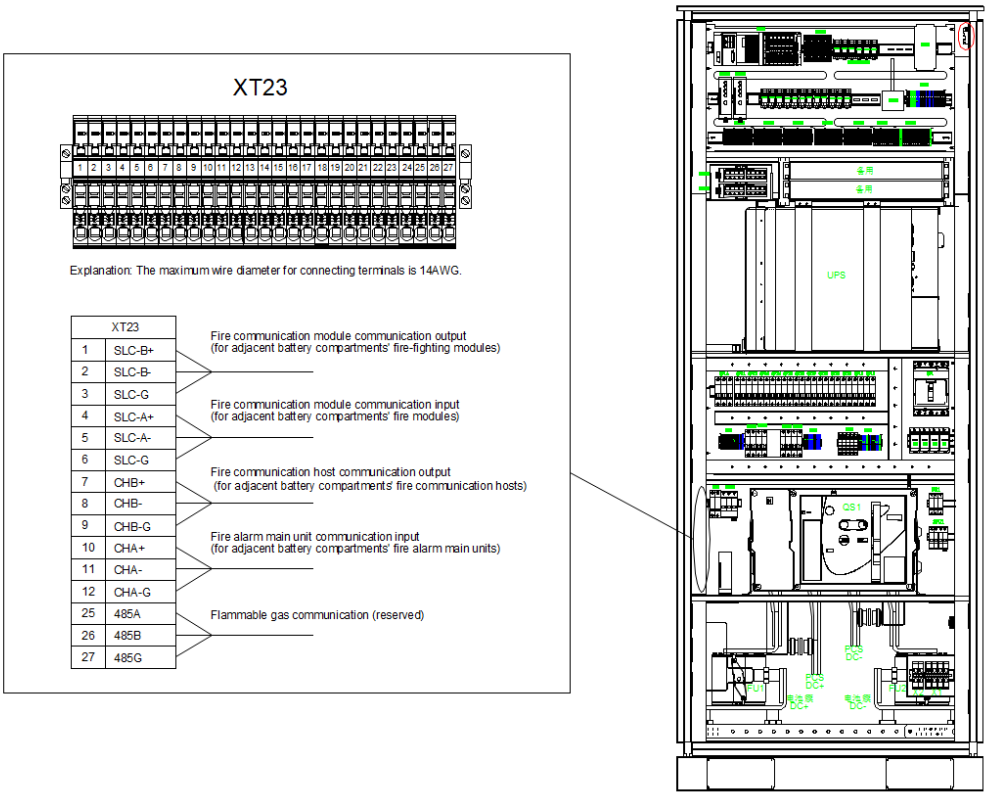


Figure 7- 9 The interface between Container and Station Level Fire Protection Host*The above figures are for reference only, please refer to the actual object received.

Terminal number	Terminal position number	Definition	Remark
XT23	1	SLC-B+	Fire communication module communication output
XT23	2	SLC-B-	
XT23	3	SLC-G	
XT23	4	SLC-A+	Fire communication module communication input
XT23	5	SLC-A-	
XT23	6	SLC-G	
XT23	7	CHB+	Fire communication host communication output
XT23	8	CHB-	

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XT23	9	CHB-G	
XT23	10	CHA+	Fire alarm main unit communication input
XT23	11	CHA-	
XT23	12	CHA-G	
XT23	25	485A	Flammable gas communication (reserved)
XT23	26	485B	
XT23	27	485G	

Operating procedure:

Step 1: Insert the fire fault, sprinkler, alarm, and communication cables into the inlet holes and enter the XT23 terminal wiring area on the left side of the cabinet.

Step 2: Connect the cables into the corresponding terminal holes of XT23 according to the definition.

Step 3: Place the cables in the left wall channel of the cabinet.

The communication interface between the Battery Container and PCS is shown in the figure below.

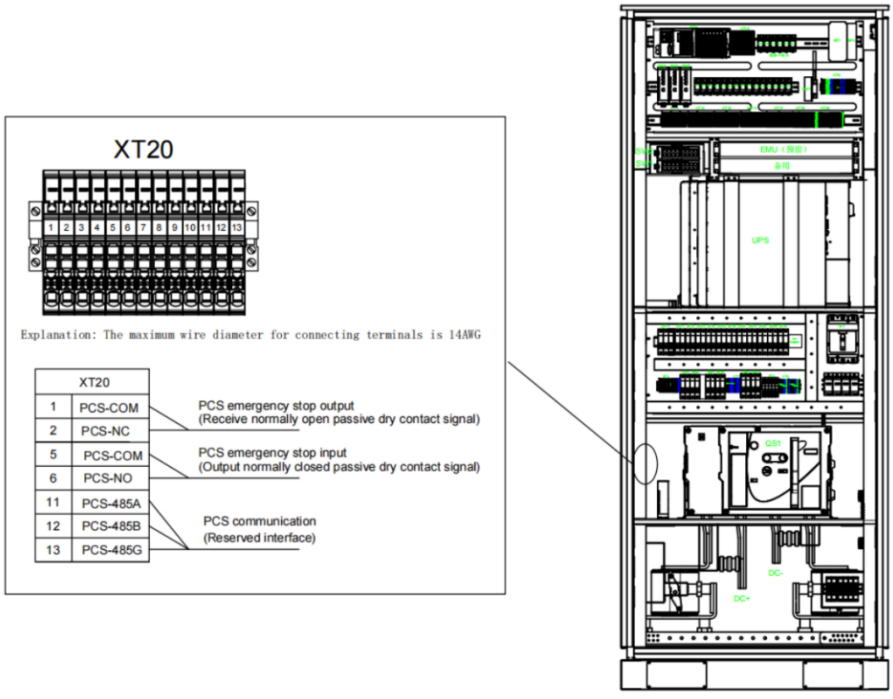


Figure 7- 10 The communication interface between the ESS and PCS
*The above pictures are for reference only, please refer to the actual object received.

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Terminal number	Terminal position number	Definition	Remark
XT20	1	PCS-COM	PCS emergency stop output (Receive normally close passive dry contact signal)
XT20	2	PCS-NC	
XT20	5	PCS-COM	PCS emergency stop input (Output normally open passive dry contact signal)
XT20	6	PCS-NO	
XT20	11	PCS-485A	PCS communication (Reserved interface)
XT20	12	PCS-485B	
XT20	13	PCS-485G	

Operating procedure:

Step 1: Insert the PCS communication, PCS emergency stop input, and PCS emergency stop output cables into the inlet holes and route them to the XT20 terminal wiring area on the left side of the cabinet.

Step 2: Connect the cables to the corresponding terminal holes of XT20 as defined.

Step 3: Arrange the cables in the left wall channel of the SCDC.

External Ethernet Communication Interfaces

The ESS container is A and B dual network communication, which can support external Ethernet communication and optical fiber communication between A and B networks.

The external Ethernet communication interface is shown in the figure below.

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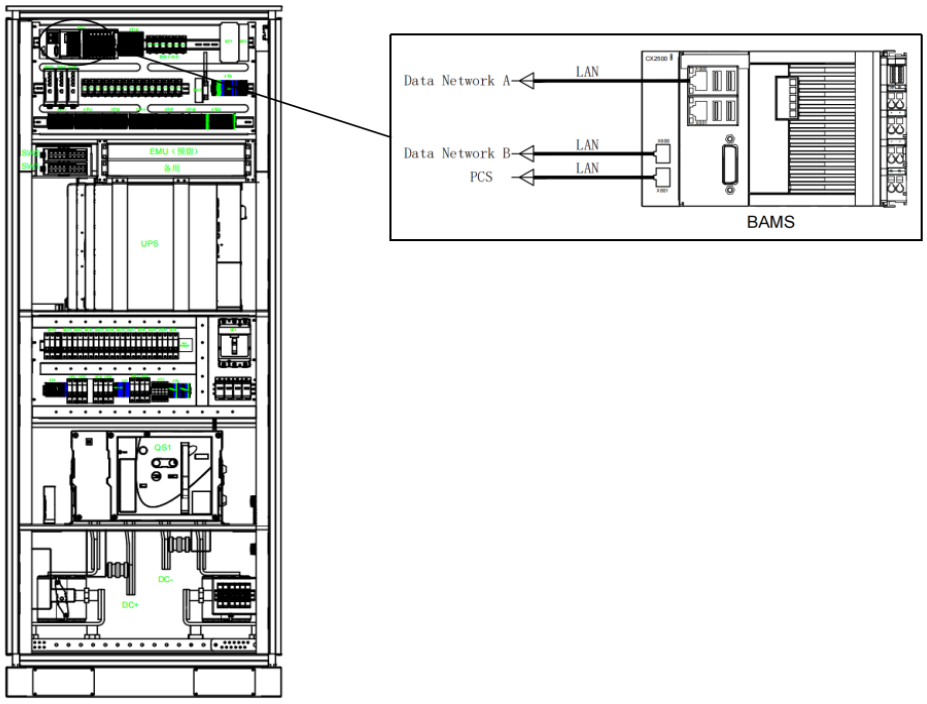


Figure 7- 11 Ethernet communication wiring location

*The above figures are for reference only, please refer to the actual object received.

Device name	Interface name	Remark
BAMU	X000	Data Network A
	X600	Data Network B
	X601	PCS

The external fiber optic communication interface is shown in the following figure

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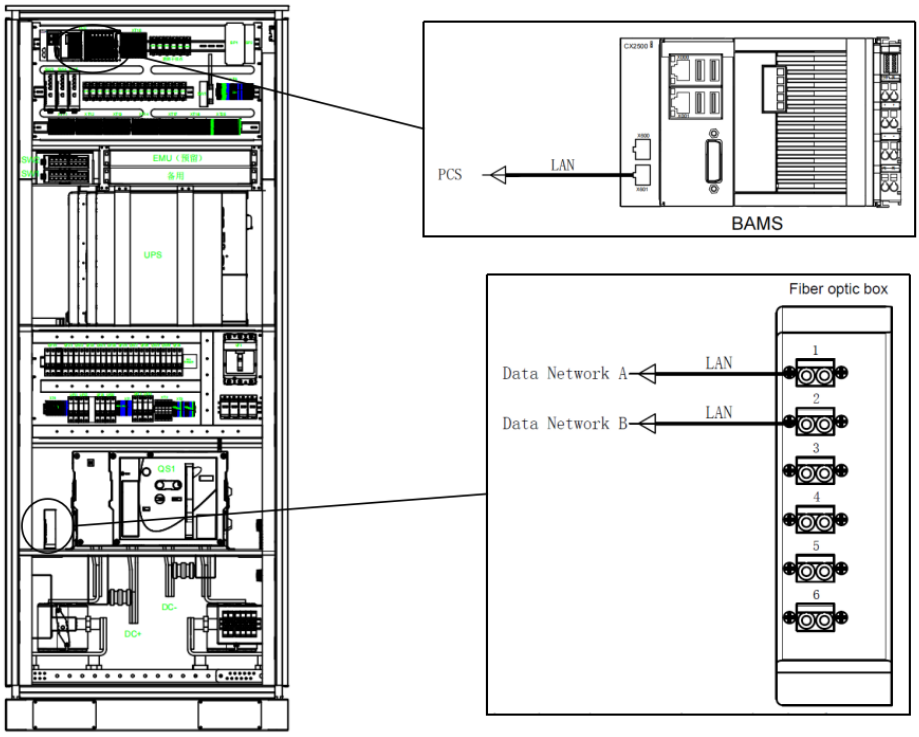


Figure 7- 12 Fiber optic communication wiring location

*The above figures are for reference only, please refer to the actual object received.

Device name	Interface name	Remark
Fiber optic box	1	Data Network A
	2	Data Network B
BAMU	X601	PCS

Communication cable requirement

It is recommended to use network cables of category 5 or above, with a maximum length of 30 meters. If the length exceeds 30 meters, an optical fiber transceiver must be configured for optical fiber communication.

(Note: The external communication interface can be made available to the customer.)

Operating procedure:

Step 1: Lead the CAT-5e cable into the inlet hole and route it to the wiring area on the left side inside the cabinet.

Step 2: Disassemble the trunking of the left side wall inside the cabinet and guide the CAT-5e cable into it.

Step 3: Connect the Ethernet port to the external device.

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7.10 UPS Cautions

CAUTION

Beware of electric shock, explosion or arcing

- Electrical equipment installation, operation, repair, and maintenance of electrical equipment must be carried out by qualified personnel.

- Wear appropriate personal protective equipment and follow electrical safety operation standards.

- Turn off all power supplies of the UPS system when operating or maintaining equipment.

- Before operating the UPS system, check for dangerous voltages across all terminals, including protective grounding.

- The UPS contains an internal power supply. Hazardous voltages may be present even when disconnected from utility power. Before installing or servicing the UPS system, ensure the unit is OFF and disconnected from the utility power and battery connections. Wait five minutes for the capacitors to discharge before turning on the UPS.

- The UPS system must be isolated from its upstream power source by a disconnecting device (disconnect, switch) that complies with local codes. The isolation device must be in a conspicuous and easily accessible location.

- The UPS must be properly grounded. Due to the possibility of a considerable leakage current, ground wires must be connected first.

Failure to follow the instructions may result in severe consequences such as personal injury or death.

CAUTION

Beware of equipment damage

- Do not install the batteries until the system is ready to be energized. The recommended time from battery installation to UPS power-up should not exceed 72 hours.

- Adhere to the charging requirements, the storage time of the battery must not

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exceed 6 months. If the UPS system remains powered off for an extended period, It is recommended to charge the battery of UPS system at least once per month for 24 hours per charge. This periodic charging helps prevent irreversible damage.

Failure to follow the instructions may result in irreversible damage to the UPS.

7.11 Power on/off Process



CAUTION

Electrical injuries, arc flash

-Operators must hold the necessary legal qualifications

- Personnel should wear personal protective equipment as required during operation;

- If any issues are identified that may affect the battery or ESS, contact after-sales support immediately. Unauthorized modifications to the internal equipment of the battery system are strictly prohibited.

- If exposed internal copper wiring or conductive material is observed, do not touch it. Notify after-sales support immediately. - In the event of other emergencies during installation, contact after-sales support immediately. Follow their guidance or wait for them to conduct on-site operations.

- If any abnormalities are observed in the storage battery system's operation, stop the system immediately and shut it down.

- For BESS with long downtime, perform a **comprehensive and meticulous inspection** to ensure all system indicators meet operational requirements before powering up.

Checking before powering on:

Before powering up, double-check the following items to ensure they are correct.

- Verify that the wiring is correct.
- Confirmed that the main circuit wiring is securely tightened.
- Ensure that the protective cover inside the equipment is firmly installed.
- Confirm that the emergency stop button is released.
- Check to ensure there are no ground faults.
- Use a multimeter to check that the AC and DC side voltages meet the startup conditions and confirm there is no overvoltage risk.

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- Check to ensure no tools or parts are left inside the unit.
- Verify that all air inlets and outlets are free of foreign objects or blockages.

Power on Process:

Steps for powering on:

Step 1: Turn the load switch on the 1#~12# PDU of the rack to "ON".

Step 2: Turn on the AC power supply main switch QF1.

Step 3: Turn on the power supply switch of the equipment in sequence:

-Turn on the power supply switch QF14 of the water cooler.

-Turn on the T1 isolation transformer power supply switch QF11 and the fire fan power supply switch QF31; or turn on the external fire independent power supply switch QF15 and the fire fan power supply switch QF32(choose one based on the actual situation).

-Turn on the T3 isolation transformer power supply switch QF13.

-Turn on the UPS output power supply switch QF21.

-Turn on the SP1 power supply switch QF22.

-Turn on the BCMU power supply switch QF23.

-Turn on the isolation switch power supply switch QF24.

-Turn on the camera power supply switch QF25.

-Turn on PCS power supply switch QF26 (backup).

-Turn on the T2 isolation transformer power supply switch QF12.

-Turn on the QF27 power supply switch for the electrical room air conditioning and control busbar cabinet.

-Turn on the dehumidifier power supply switch QF28.

-Turn on the backup power supply switch QF29

Power on operation completed.

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Power off Process:

Steps for powering off:

Step 1: Turn off the power supply switch of the equipment in sequence.

–Disconnect the power supply switch QF14 of the water cooler.

–Disconnect the T1 isolation transformer power supply switch QF11 and the fire fan power supply switch QF31; or disconnect the external independent fire power supply switch QF15 and the fire fan power supply switch QF32(choose one based on the actual situation).

-Disconnect the T3 isolation transformer power supply switch QF13.

-Disconnect the UPS output power supply switch QF21.

-Disconnect the SP1 power supply switch QF22.

-Disconnect the BCMU power supply switch QF23.

-Disconnect the isolation switch power supply switch QF24.

-Disconnect the camera power switch QF25.

-Disconnect the PCS power supply switch QF26 (backup).

-Disconnect the T2 isolation transformer power supply switch QF12.

-Disconnect the QF27 power supply switch for the electrical room air conditioning and control busbar cabinet.

-Disconnect the dehumidifier power supply switch QF28.

-Disconnect the backup power supply switch QF29.

Step 2: Turn off the AC power supply main switch QF1.

Step 3: Turn the load switch on the 1#~12# PDU of the rack to "OFF".

Power-off operation completed.

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8 HMI

This section organizes and elaborates on the touchscreen program structure of the ESS control system. It includes primary data display and essential function operation to assist developers and O&M personnel in understanding the system. The goal is to facilitate more effective system maintenance and operation.

8.1 Touchscreen Content and Structure

The ESS touchscreen program consists of six key sections:

System operation information

Alarm information

Event recording

System diagram

User management

System settings

8.2 Touchscreen Program

The touchscreen provides information about the operation of the main equipment of the energy storage control system and facilitates the issuance of control instructions to the system. The connection between the ESS controller and the touchscreen is established via a network cable. The ESS controller transmits the real-time information of each equipment to the touchscreen; the touchscreen sends control instructions to the ESS controller, which then communicates them to the appropriate equipment.

8.3 Basic Environment Requirements for Touchscreen

Equipment Environment Requirements

Requirement Items		**Parameters**
Altitude (m)		≤3000m
Installation Location		Inside an outdoor container
Operating Ambient Temperature		-20~45°C
Seismic resistance	Horizontal acceleration	0.2
	Vertical acceleration	0.1

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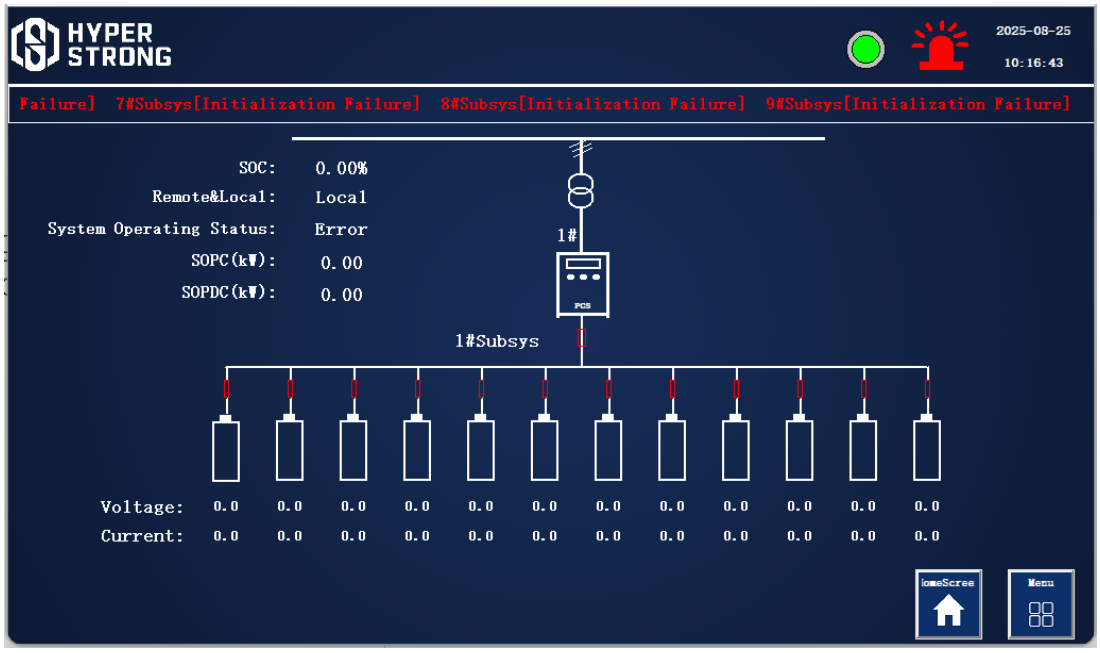
Relative air humidity	The average daily relative humidity should be ≤ 95%. The average monthly relative humidity should be ≤ 90%. No condensation on the equipment.

8.4 Touchscreen Basic Functions

Touchscreen Main Page

The main page of the touchscreen serves as the central hub for system operation and monitoring. It provides:

- A real-time overview of the system's performance.
- Quick access to critical features such as alarms, events, and system configurations.
- Easy navigation to other sections for detailed information or adjustments.



Once the program is in operation, the first screen displayed is the main page, which provides key information about the ESS control system. This information includes

Subsystem SOC,

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Remote local state,

Working state,

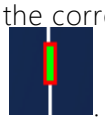
Maximum charging power and discharging power.

Voltage and current of the rack.

The **topology diagram**, centrally located on the main interface, visually represents the control structure of the entire ESS. Key elements of the topology diagram include: SCDC and relay switches state:

- When the interface icon is shows  , it indicates that at least one relay among the main positive and main negative relays of the battery cluster is disconnected.

- When the icon changes to  , it indicates that both the main positive and main negative relays are closed.

- When all the relay switches of the subsystem are closed. the corresponding busbar will be closed, and the icon  will change to  .

- The icon  indicates the basic state of the SOC of the rack,

  At the top right of the main interface, the following additional information is displayed:

- Communication state of the PLC,

- Alarm display of the ESS

- Current date and time

  aA real-time alarm scrolling display of the ESS is located in the center of the main interface to ensure that critical alerts are prominently visible.

Navigation:

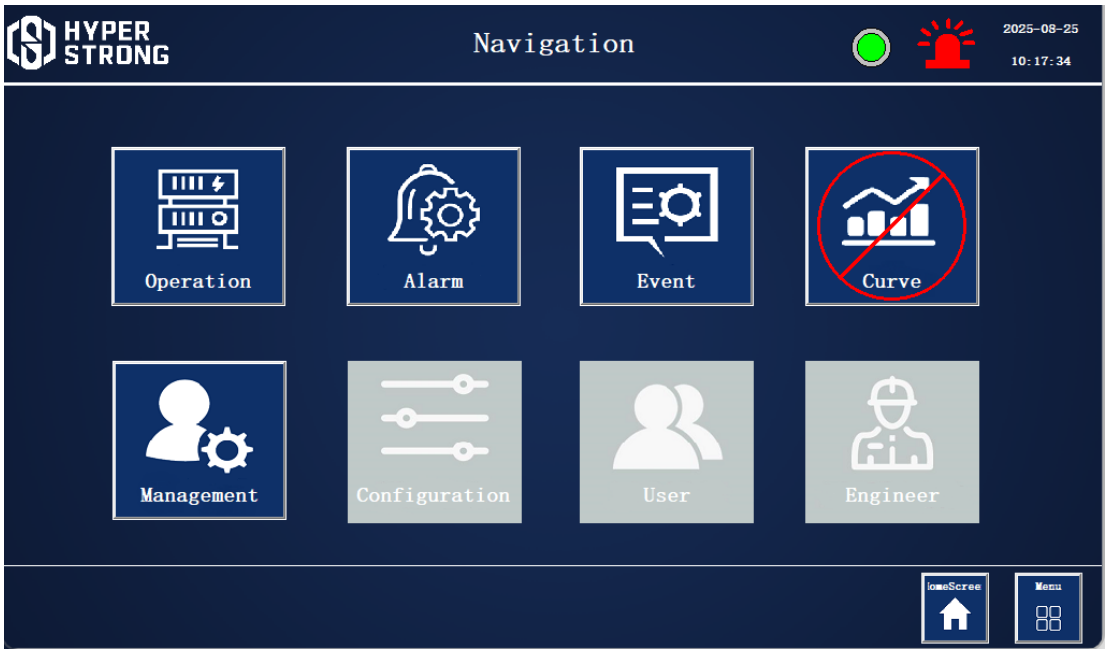
At the bottom right corner of the main interface, users can switch between the **Home Screen** and **Navigation Screen using** 

or 

buttons.

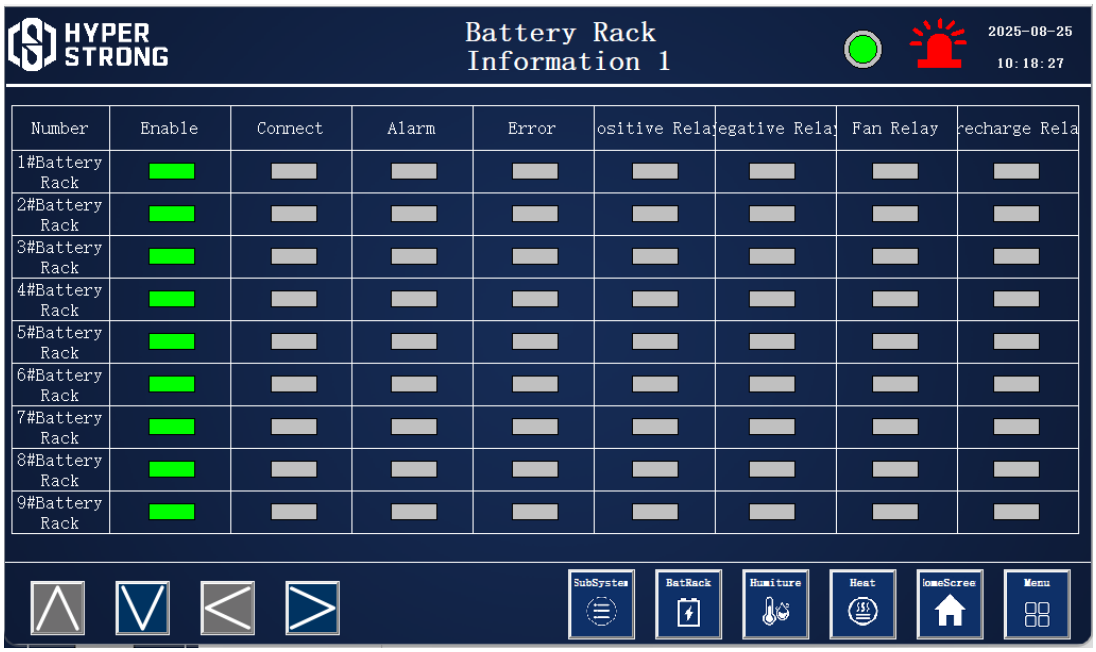
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Rack Information Interface

When the user clicks on "Operation" from the main page, the system transitions to the Rack Information Interface. This interface displays detailed operational data for each rack in the BCMU.

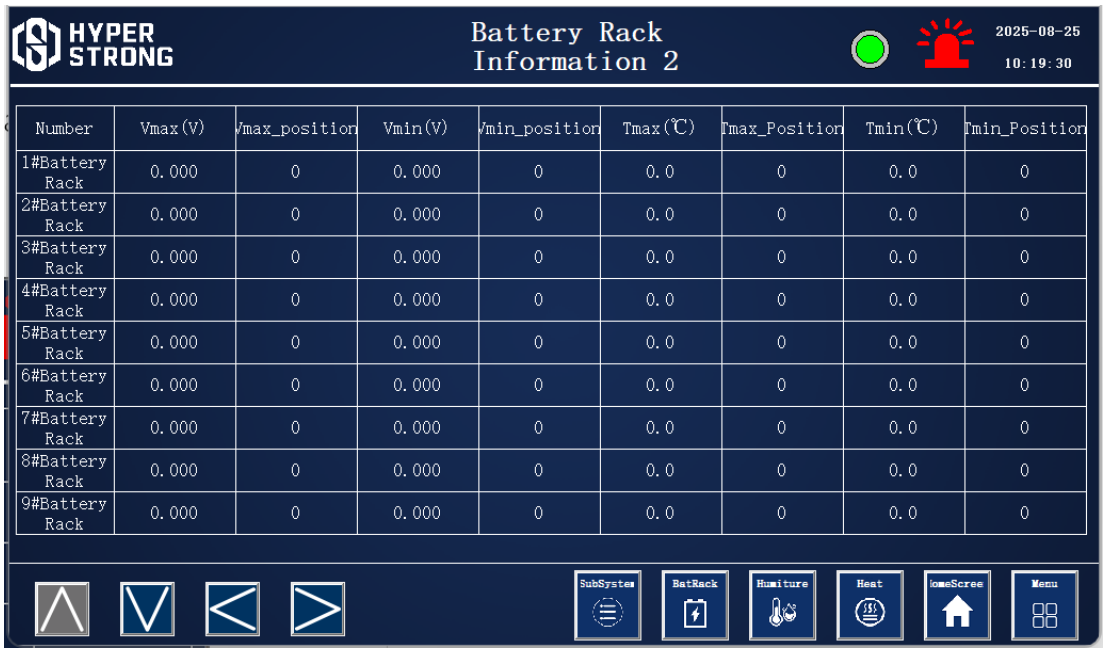


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Entering the rack information interface, operators will see the current status of the racks, including “Enable”, “Connect”, “Alarm”, “Error”, “Positive Relay”, “Negative Relay”, “Fan Relay”, and “Pre-charge Relay”.

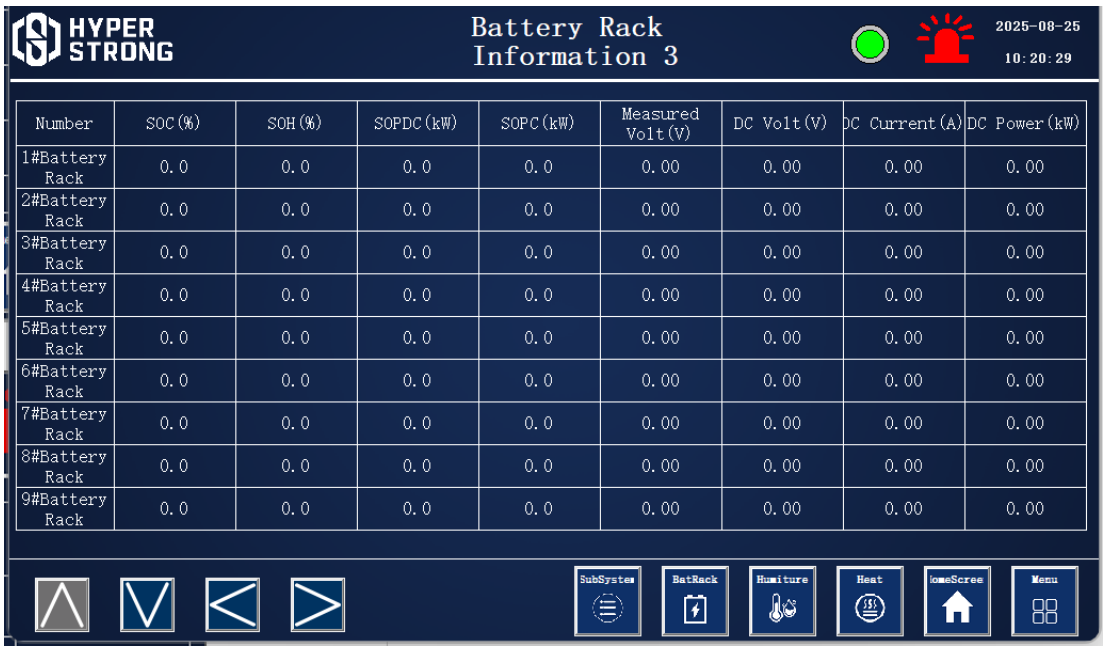
Click on  , it will enter Battery Cluster Information 2, Battery Cluster Information 3, etc.

Click  to return to the return to the previous page. Operators can click  and  to enter any interface to check the operation information of each cluster of batteries.



Number	Vmax(V)	Vmax_position	Vmin(V)	Vmin_position	Tmax(℃)	Tmax_Position	Tmin(℃)	Tmin_Position
1#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
2#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
3#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
4#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
5#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
6#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
7#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
8#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
9#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0

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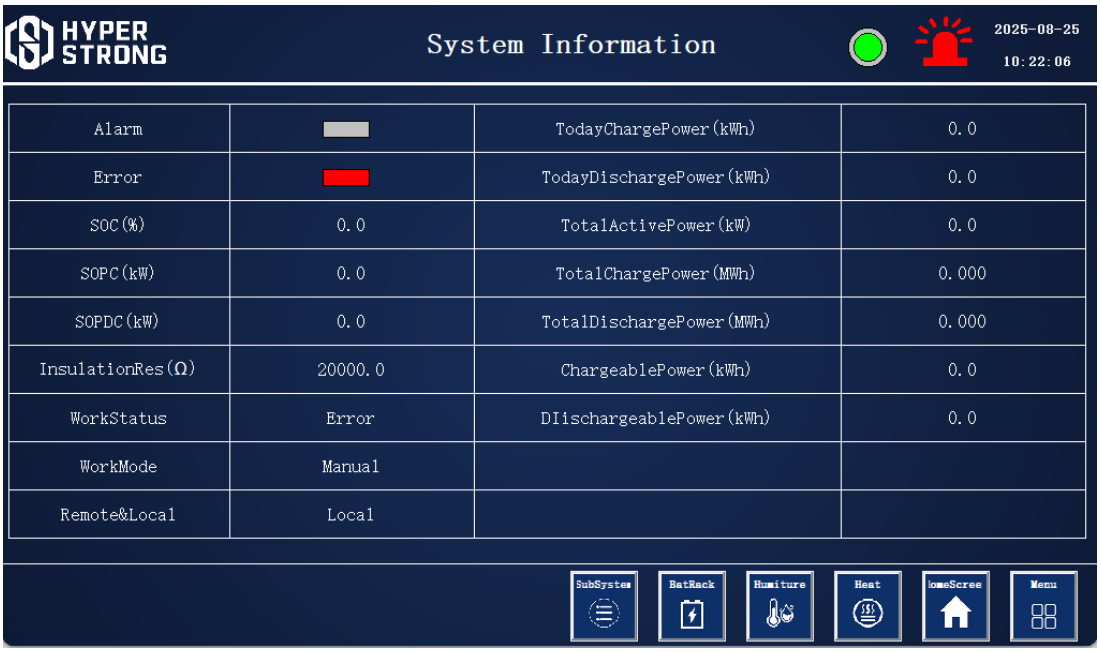


Number	SOC(%)	SOH(%)	SOPDC(kW)	SOPC(kW)	Measured Volt(V)	DC Volt(V)	DC Current(A)	DC Power(kW)
1#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
2#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
3#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
4#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
5#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
6#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
7#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
8#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
9#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00

Click 

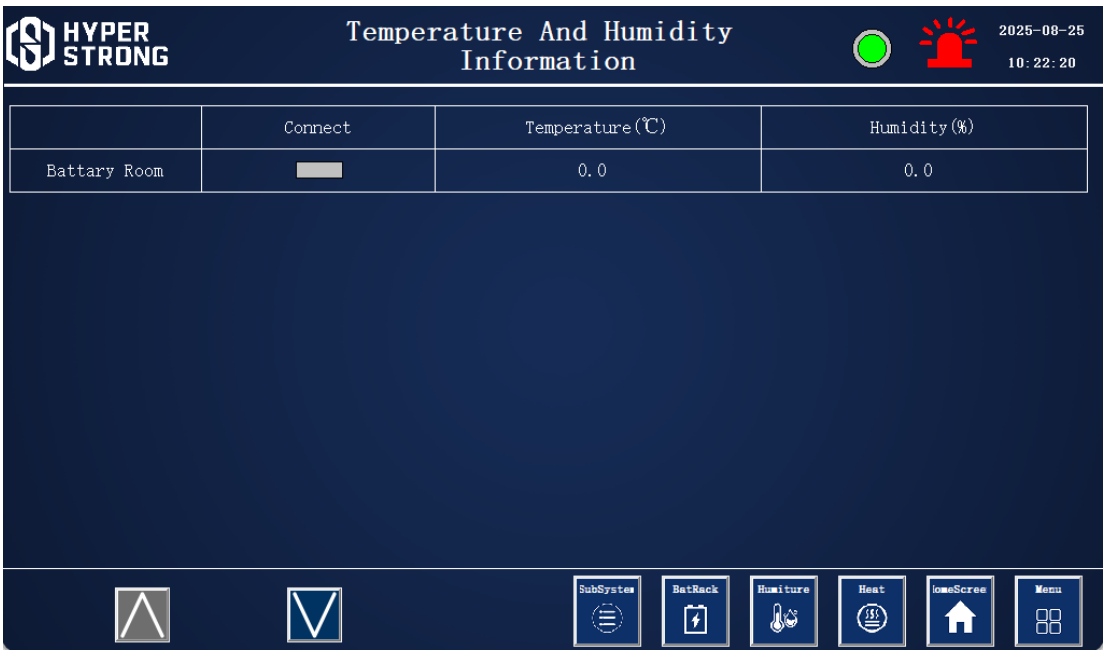
under the battery cluster information will navigate to the System Information Interface; while clicking 

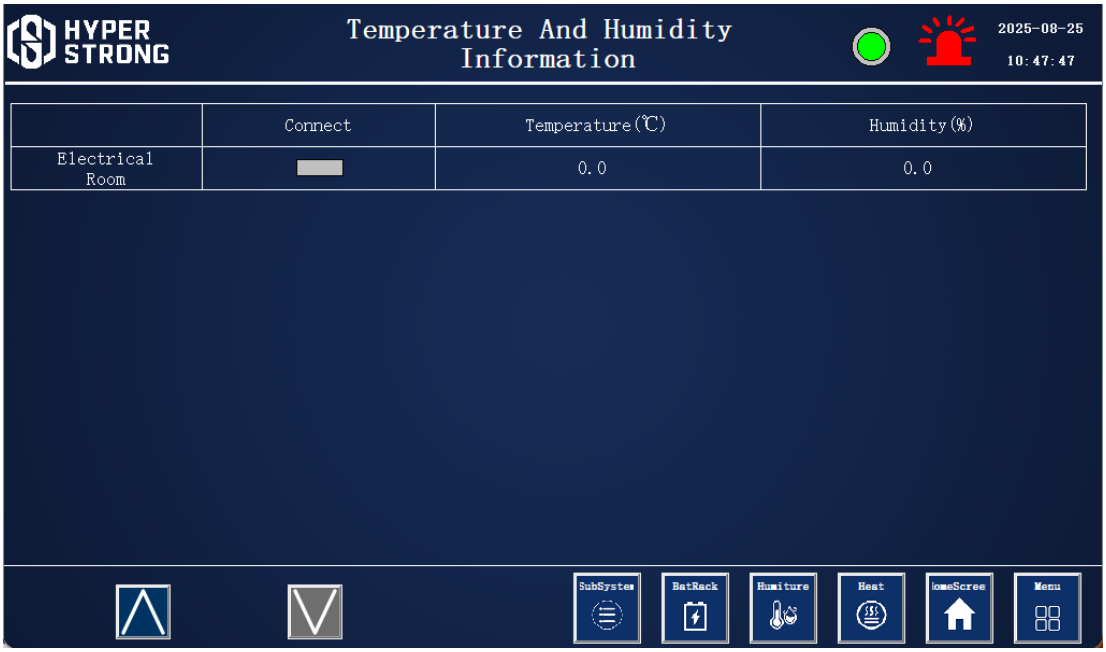
will lead to the Temperature & Humidity Information Interface. Operators can also click to review the operation information of each part of the BMS.



Alarm		TodayChargePower(kWh)	0.0
Error		TodayDischargePower(kWh)	0.0
SOC(%)	0.0	TotalActivePower(kW)	0.0
SOPC(kW)	0.0	TotalChargePower(MWh)	0.000
SOPDC(kW)	0.0	TotalDischargePower(MWh)	0.000
InsulationRes(Ω)	20000.0	ChargeablePower(kWh)	0.0
WorkStatus	Error	DIischargeablePower(kWh)	0.0
WorkMode	Manual		
Remote&Local	Local		

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Alarm Interface

In the main interface or other interfaces, click 

to enter the menu. Click "Alarm" to enter the real-time alarm interface and check real-time alarm. Click 

to enter the historical alarm information interface and check historical data. Click 

to enter the real-time alarm information interface and check real-time alarms.

The main alarm interface information consists of two sections:

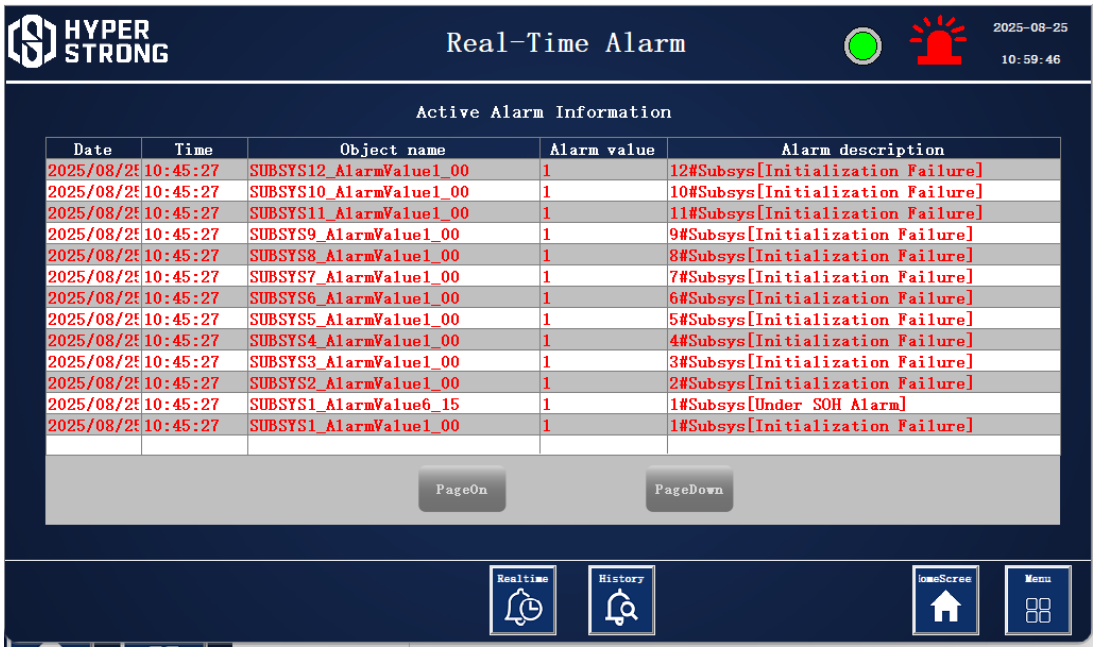
1.Real-time and a historical alarm information display (the alarm information in the picture is only for reference).

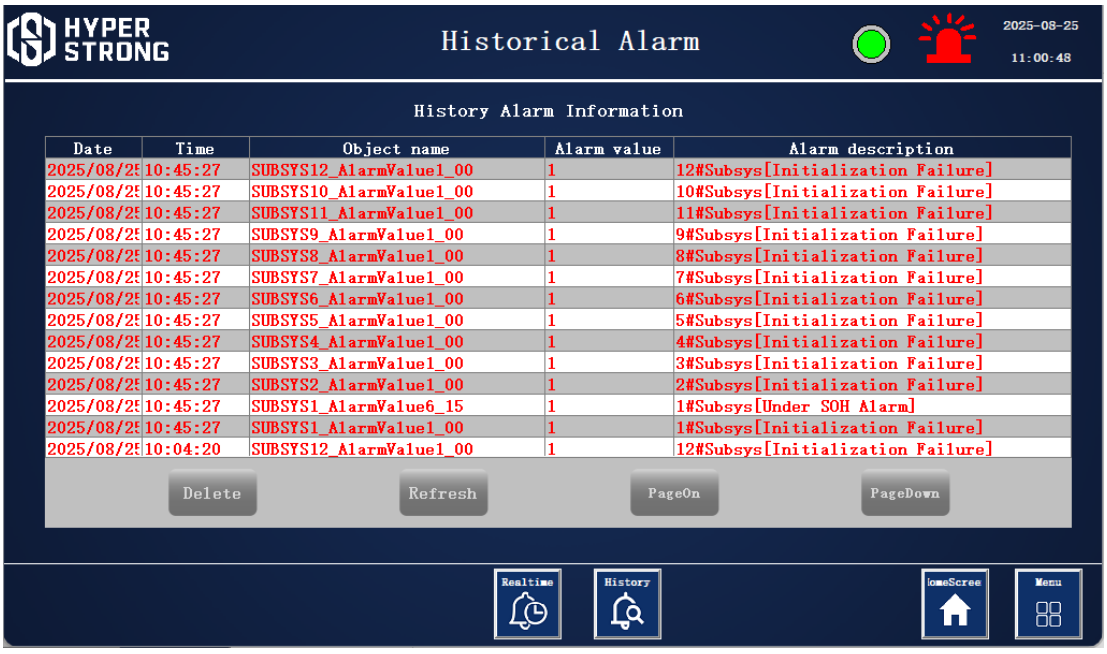
2. Historical Alarm information Display

The real-time alarm interface can show up to 14 alarm messages at a time. Users can navigate through additional alarm messages by flipping up or down the page.

The historical alarm interface can also display up to 14 alarm messages at a time. Operator can perform operations such as deleting historical alarm messages, refreshing historical alarm messages, and navigating through messages by flipping up or down.

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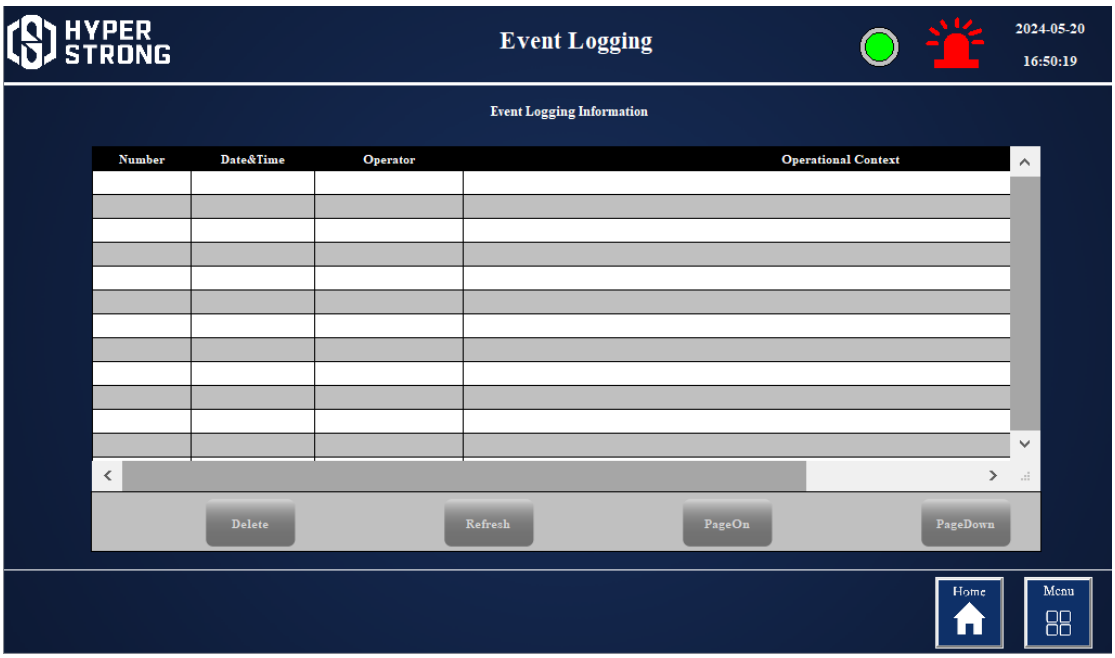
Event Logging Interface

Click 

on the main page or other pages to access the Menu. Click “Event Logging” to enter Event Logging section and review operation records.

The event logging interface includes the Date & Time, Operator, and Operational Context. Additionally, operators have the option to delete records, refresh the page, and navigate up and down the page.

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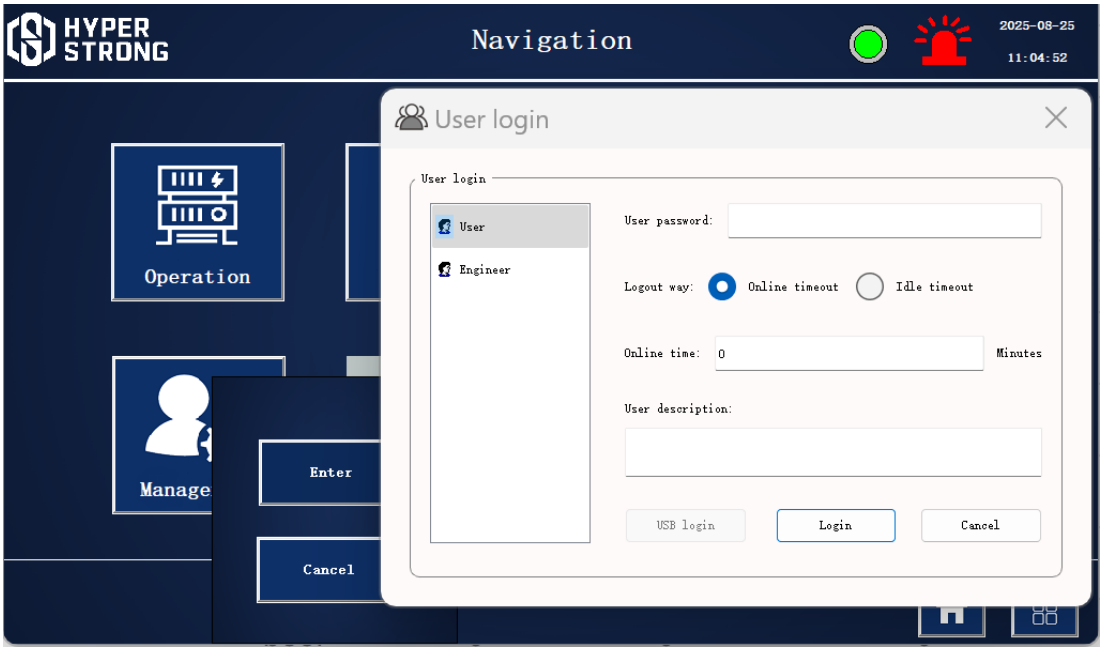
User Management Interface

Click 

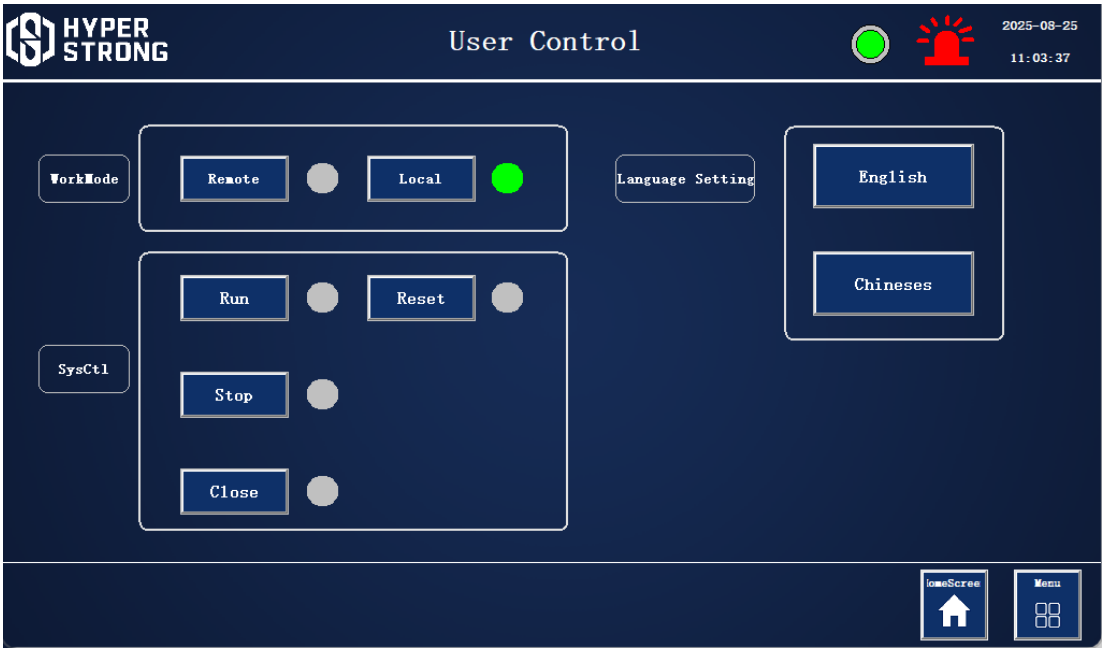
, enter the Menu page and click “User Management,” and the Login interface will pop up in the main interface or other interfaces. Select “User” or “Engineer”, enter the password, and click “Login” to complete the login. If you log in as an engineer, the “Engineer Operation” button in the menu interface becomes operational.



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Click on “User” to enter User Control. This interface's main functions include subsystem run/shutdown/reset control, remote/local mode switching, and subsystem selection/ operation. (Contact the HyperStrong’s operation and maintenance personnel to obtain the password.)



The user can select the mode based on the situation and click RUN depending on the operation condition; then, the corresponding subsystem will turn to the running state. Click Stop, the corresponding subsystem will be shut down. Click Close the

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corresponding subsystem will be turn off. If a subsystem enters a fault-recoverable state and needs to reset, click 

, and the system reset instruction will be successfully issued.

8.5 Communications

The HMI has a network cable interface and connects to BAMU-PLC via a network cable. Modbus TCP/IP transmission protocol is used for data transmission.

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9 Emergency Operation and Accident Handling

The faults of the ESS primarily include battery faults, control system faults, equipment faults and physical damage. For common faults, refer to the maintenance manual. This chapter addresses emergency faults, such as fire and electric leakage.

If there is any uncertainty about the identifying or handling of abnormal conditions, properly contact the relevant technical or after-sales service department of our company for professional technical support.

When contacting us, providing the following information will help locate and resolve the cause of the fault more quickly and accurately:

- Equipment serial number, production date and software version
- Fault information and brief description
- Photos of faulty parts
- Information about the manufacturer, model, and configuration of the external devices connected to the energy storage system (e.g. transformers, upstream circuit breakers, and cable models).

9.1 Emergency Handling

If any of the following conditions (including but not limited to) occur, immediately stop using the ESS container and the adjacent battery container to prevent electric shock, fire, or explosion.

In Case of Fire:

In case of electrical equipment catches fire, immediately cut off the power supply and use dry powder or carbon dioxide fire extinguishers to extinguish the fire.

If there is an open flame outside the ESS container, immediately move away from the ESS system and contact the local fire emergency services for assistance.

After emergency troubleshooting, stand and observe for a specific period. If no abnormality occurs, proceed to the next step of troubleshooting.

Typical ESS emergencies and anomalies:

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- Abnormal sound, spark, odor or other irregularities occurs during the operation or maintenance.
- Smoke or fire in the container.
- Expansion, rupture, melting, deformation of the battery module.
- External or internal short circuit.
- Extreme weather conditions, such as rainstorms and flash floods.
- Thermal runaway occurs during use, indicated by a battery temperature exceeding 85°C.
- Structural damage **that has not yet caused a severe accident**.
- The coolant of the chiller leaks.
- The auxiliary power supply is abnormal or the communication is interrupted.
- Damage to the main structure of the battery container or IP protection of the battery system.
- Loose or burnt marks on electrical connections.
- Insulation fault alarm. If triggered, stop using immediately and verify whether the alarm is false.

If any the above situation occurs (not limited to the above case), resulting in the ESS's dysfunction, contact our professional maintenance team or technical support immediately.

If the situation is critical, activate the contingency plan (refer to the maintenance manual), and notify first responders or fire services.

CAUTION

Before troubleshooting faults, replacing spare parts, or upgrading software, ensure that the DC and AC power supplies are disconnected. Use a multimeter to check that all devices are powered off. Ensure that the devices under maintenance cannot unexpectedly power on.

9.2 Emergency Operations

Power-off process

① Power off via remote control

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The BAMU sends a shutdown command to BMS after receiving the shutdown command from EMS. Once powered off, the system reverts to its initial state. The ESS container main circuit disconnects, **eliminating high voltage in the main circuit.** The RUN indicator on the cabinet door turns off after the system is powered off.

② Power off via local control

Open the side door of the container, manually toggle the button switch to the OFF state. Confirm that the green indicator on the panel of the BPU is off.

Emergency shutdown operation process

① Press the E-stop button on the ESS container or SCDC. (Refer to the ESS container's appearance in Section 5.2 and the SCDC's appearance in Section 5.8 for details.) When the SCDC is in the local control mode, it can be powered off using the HMI. Refer to the alarm information in section 8.4.3 and the user operation interface in section 8.4.5.

② When the SCDC is in the remote control mode, the control commands can be issued by the EMS. Local operation using the HMI is disabled. Detailed operation steps are provided in the EMS manual.

CAUTION

Manual emergency stop must not be performed in event of fire.

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10 Product Labels

The images below are for reference only. Some are examples for illustration purposes. Refer to the actual object received.



Figure 10-1 ES label

(Remark: a circular green reflector sign with a minimum diameter of 100 mm, featuring the letters "ES" on)



Figure 10-2 Restricted access sign



Figure 10-3 Typical restricted access sign

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Figure 10-4 Explosion hazard label
(Remark: Minimum dimensions of 175mm*175mm)



Figure 10-5 Toxic fumes hazard label



Figure 10- 6 Shutdown procedure notice label

Note:

Disconnection devices must be labelled with an identification name or number as specified in the BESS wiring diagram, consistent with the shutdown procedure.

The minimum size of this sign must ensure readability from a distance of 1m.

All switches shall clearly and reliably indicate the isolating position of the device. The symbols "0" (off) and "I"(on) fulfil this requirement.
The isolation switch model: MVS 25 DA1 A4F2.

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11 Contact Information

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Long-term storage